

Number System

Binary

Octal

Hexa

Decimal

• Decimal to Binary

$$(114.35)_{10} = (?)_2$$

2	114	
2	57	0
2	28	1
2	14	0
2	7	0
2	3	1
2	1	1
	0	1

	$\times 2$
0	.70
	$\times 2$
1	.40
	$\times 2$
0	.80
	$\times 2$
1	.60
	$\times 2$
1	.20

01011

$$\rightarrow (1110010.01011)_2$$

• Decimal to Octal

$$(85.75)_{10}$$

$$\begin{array}{r} 8 \overline{) 85} \\ \underline{80} \\ 5 \\ 8 \overline{) 10 - 5} \\ \underline{8} \\ 2 \\ 8 \overline{) 1 - 2} \\ \underline{0} \\ 1 \end{array} \quad \uparrow$$

$$\begin{array}{r} .75 \\ \times 8 \\ \hline 6.00 \end{array}$$

→ 125

• Decimal to Hexa.

$$(285.75)_{10}$$

$$\begin{array}{r} 16 \overline{) 285} \\ \underline{160} \\ 125 \\ 16 \overline{) 125} \\ \underline{112} \\ 13 \\ 16 \overline{) 13} \\ \underline{0} \\ 13 \end{array} \quad \uparrow$$

$$\begin{array}{r} .75 \\ \times 16 \\ \hline 12.00 \\ \end{array}$$

→ 11D

$$* (50.25)_{10} \rightarrow (?)_5$$

$$\begin{array}{r} 5 \overline{) 50} \\ \underline{50} \\ 0 \\ 5 \overline{) 10 - 0} \\ \underline{10} \\ 0 \\ 5 \overline{) 2 - 0} \\ \underline{0} \\ 2 \end{array}$$

$$\begin{array}{r} .25 \\ \times 5 \\ \hline 1.25 \\ \\ .25 \\ \times 5 \\ \hline 1.25 \\ \\ .25 \\ \times 5 \\ \hline 1.25 \end{array}$$

→ 200.111.

• Binary to Decimal

$$(1110010.010)_2$$

$$\rightarrow 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3}$$

$$\rightarrow 64 + 32 + 16 + 0 + 0 + 2 + 0 + 0 + \frac{1}{4} + 0$$

$$\rightarrow (114.25)_{10}$$

• Octal to Decimal

$$(734.35)_8$$

$$\rightarrow 7 \times 8^2 + 3 \times 8^1 + 4 \times 8^0 + 3 \times 8^{-1} + 5 \times 8^{-2}$$

$$\rightarrow (476.375)_{10}$$

• Hexa to Decimal

$$(AB.D2)_{16}$$

$$\rightarrow 10 \times 16^1 + 11 \times 16^0 + 13 \times 16^{-1} + 2 \times 16^{-2}$$

$$\rightarrow 171.8203$$

• Hexa to Oct

$$(3FA.4B)_{16}$$

Bin $\begin{array}{|c|c|c|c|c|c|c|c|} \hline 0011 & 1111 & 1010 & 0100 & 1011 & 0 & & \\ \hline \end{array}$

Oct $\rightarrow (177.226)_8$

$$(FG.AD)_{16}$$

Bin $\rightarrow 01111101101011010$

Oct $\rightarrow (366.532)_8$

• Addition

$$\begin{array}{r} 1101.11 \\ + 101.01 \\ \hline 10011.00 \end{array}$$

• Subtraction

$$\begin{array}{r} 1101 \\ - 0111 \\ \hline 0110 \end{array}$$

1's Complement & 2's Complement

Q * $(59)_{10} - (25)_{10}$

$59 \rightarrow 00111011$

$25 \rightarrow 00011001$

11100110

+1

$-25 \rightarrow 11100111$

$59 \rightarrow 00111011$

$-25 \rightarrow 11100111$

100100010

↓
34

2's complement →

1's complement + 1

Sign bit, 0 → (+)

" 1 → (-)

$$(-55) + (-30)$$

$$* \underline{-55 - 30}$$

64 32 16 8 4 2 1

1 1 0 1 1 1

55

30

$$55 \rightarrow 00110111$$

$$11001000$$

+1

$$(-55) \quad 11001001 \leftarrow 2's \text{ complement}$$

$$30 \rightarrow 00011110$$

$$11100001$$

+1

$$(-30) \quad 11100010 \leftarrow 2's$$

Now,

$$(-55) \rightarrow 11001001 \rightarrow -55$$

$$+ \quad (-30) \rightarrow 11100010 \rightarrow -30$$

$$\begin{array}{r} 11001001 \\ + 11100010 \\ \hline 1 \boxed{1} 0101011 \end{array}$$

① (-ve)

$$-85 \quad \text{Sign bit (1)} \quad \text{①}$$

Result (-ve)

$$10101011$$

$$01010100$$

+1

$$\begin{array}{r} 10101011 \\ + 01010100 \\ + 1 \\ \hline 01010101 \end{array}$$

Ans: -85

• R's Complement

• (R-1)'s Complement

Decimal

10's

9's

Binary

2's

1's

Octal

8's

7's

Hexa

16's

F's

Q • (543)₁₀

Given,

R = 10

n = 3

N = 543

$R^n - N$

R = Radix or Base

n = Number of digits

N = Given Number

→ find the R's complement :

$R^n - N$

$= 10^3 - 543$

$= 1000 - 543$

$= 457$

Ans

Trick : (for all)

• $(XXXX)_8 - 1264$

Q • (1264)₈ find 8's complement

$= (7777)_8 - 1264$

$= 6513$ (7's complement)

so,

$= 6513 + 1$

$= 6514$

Ans

→ [+1 টা সবগুলোতে করে,]

[এখানে যেহেতু ৪(৬) কক্ষ আছে

তার মানে Octal, তাই Octal

এর Highest Number ৭ হবে,

যদি হলে ৭, (X=7), Decimal এর

Highest হলে ১, তাই ১ হবে, Binary এর Highest 1, (X=1) হবে, সবগুলোতে করে চাও

Find R-1's Complement :

$$(543)_{10}$$

$$= (xxx)_{10} - 543$$

$$= (999)_{10} - 543$$

$$= 456$$

Ans

{ Same trick as R's complement
- किंतु एकांत +1 करता है ना }

• Boolean logic :

$$1 \cdot A = A$$

$$1 + A = 1$$

$$A + A = A$$

$$A \cdot A = A$$

$$A \cdot \bar{A} = 0$$

$$A + \bar{A} = 1$$

• De Morgan's law

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

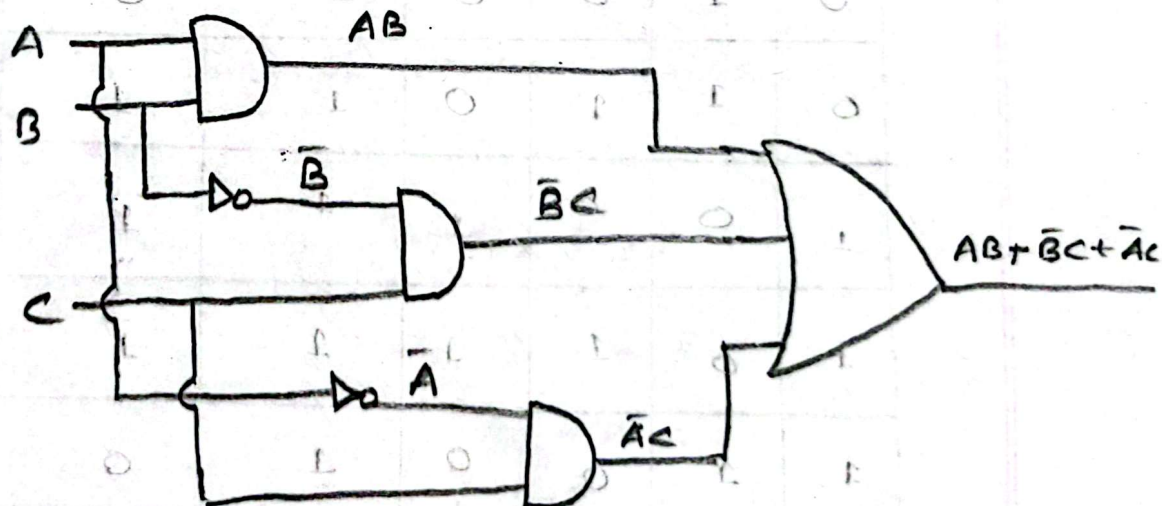
$$\overline{A + B + C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

$$\overline{A \cdot B \cdot C} = \bar{A} + \bar{B} + \bar{C}$$

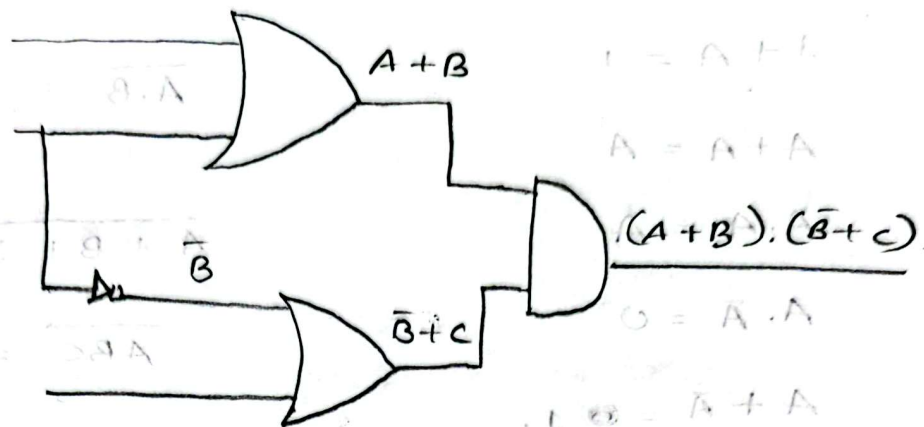
• Distributive law

$$A + BC = (A + B)(A + C)$$

$$* AB + \bar{B}C + \bar{A}C$$



* $(A+B)(\bar{B}+C)$



• Truth table :

A	B	C	$\bar{B}$	$A+B$	$\bar{B}+C$	$(A+B)(\bar{B}+C)$
0	0	0	1	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	1	0	1	1	1
1	0	0	1	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	0	0
1	1	1	0	1	1	1

$$A\bar{B} + AC + B\bar{B} + BC$$

$$\Rightarrow A\bar{B} + AC + BC \rightarrow \text{SOP (Sum of product)}$$

$$\Rightarrow (A+B)(\bar{B}+C) \rightarrow \text{POS (Product of sum)}$$

SOP

POS

When function $F=1$

$$A \rightarrow 1$$

$$\bar{A} \rightarrow 0$$

when function $f=0$

$$A \rightarrow 0$$

$$\bar{A} \rightarrow 1$$

Eq. $\frac{1-00m_0}{\bar{A}\bar{B}}, \frac{01m_2}{AB}$

Eq. $00, 01$

C.S. $A+B, A+\bar{B}$

* Express f in sum of product form;

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

when $F=1$

$$0 \rightarrow \bar{A}$$

$$1 \rightarrow A$$

$$F = \bar{A}B + A\bar{B} + AB$$

$$(1+1)00 + (1+0)01 + 10A = 11$$

$$(0\bar{A} + 0\bar{A}) + (F(A,B) = m_1 + m_2 + m_3)$$

$$\Rightarrow F(A, B) = \sum m(1, 2, 3)$$

$$\star F(A, B) = \sum m(1, 2, 3)$$

$$F = \bar{A}B + A\bar{B} + AB$$

Standard or

canonical form

$$= \bar{A}B + AB + A\bar{B} + AB$$

$$= B(\bar{A} + A) + A(\bar{B} + B)$$

$$= A + B$$

[Minimal SOP]

• $A + B$ standard (Minimal SOP to canonical form)

$$\bar{A} + A, 1 + B, 1$$

$$\Rightarrow A(B + \bar{B}) + B(A + \bar{A})$$

$$\Rightarrow AB + A\bar{B} + AB + \bar{A}B$$

$$\Rightarrow A\bar{B} + \bar{A}B + AB$$

$$\star F(A, B, C) = A + BC$$

$$= A.1.1 + BC.1$$

$$= A.(B + \bar{B}).(C + \bar{C}) + BC.(A + \bar{A})$$

$$= (AB + A\bar{B})(C + \bar{C}) + (ABC + \bar{A}BC)$$

$$= (ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC + \bar{A}BC) *$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC$$

111

110

101

100

011

↓
7

↓
6

↓
5

↓
4

↓
3

$$\therefore f(A, B, C) = \sum m(3, 4, 5, 6, 7)$$

• POS (Product of Sum)

A	B	F
0	0	0
0	1	1
1	0	0
1	1	1

When $F=0$, $A \rightarrow 0$

$\bar{A} \rightarrow 1$

$$F = (A+B) \cdot (\bar{A}B)$$

$$\square \text{ Maxterm} = \pi M(M_0, M_2)$$

$$\square (A+B) \cdot (\bar{A}B) \rightarrow \text{Standard POS.}$$

$$= A\bar{A} + AB + \bar{A}B + B$$

$$= AB + \bar{A}B + B$$

$$= B(A\bar{A}) + B$$

$$= B+B$$

$$= B \rightarrow \text{Minimal POS.}$$

Maisha_3AF

$$\begin{aligned}
 * F(A, B, C) &= (A + \bar{B}) (A + C) \\
 &= (A + \bar{B} + 0) (A + C + 0) \\
 &= (A + \bar{B} + C \cdot \bar{C}) (A + C + B \cdot \bar{B}) \\
 &= (A + \bar{B} + C) (A + \bar{B} + \bar{C}) (A + B + C) (A + \bar{B} + C) \\
 &= [A + BC = (A + B) (A + C)]
 \end{aligned}$$

$$\begin{aligned}
 &= (A + \bar{B} + C) (A + \bar{B} + \bar{C}) (A + B + C) \\
 &\quad (010) \quad (011) \quad (000)
 \end{aligned}$$

↓ 2	↓ 3	↓ 0
0	0	0
1	0	0
0	1	0
0	1	0
0	0	1
0	0	1
0	0	1

$$\therefore \text{Maxterm} = \pi M (M_0, M_2, M_3)$$

$$0 \square (A + \bar{B}) (A + C) \rightarrow \text{SOP}$$

$$= A + AC + A\bar{B} + \bar{B}C$$

$$= A(1 + C) + A\bar{B} + \bar{B}C$$

$$= A + A\bar{B} + \bar{B}C$$

$$= A \cdot 1 \cdot 1 + A\bar{B} \cdot 1 + \bar{B} \cdot C \cdot 1$$

$$= A \cdot (B + \bar{B}) \cdot (C + \bar{C}) + A\bar{B} \cdot (C + \bar{C}) + \bar{B}C(A + \bar{A})$$

$$= A(BC + B\bar{C} + \bar{B}C + \bar{B}\bar{C}) + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}\bar{C}$$

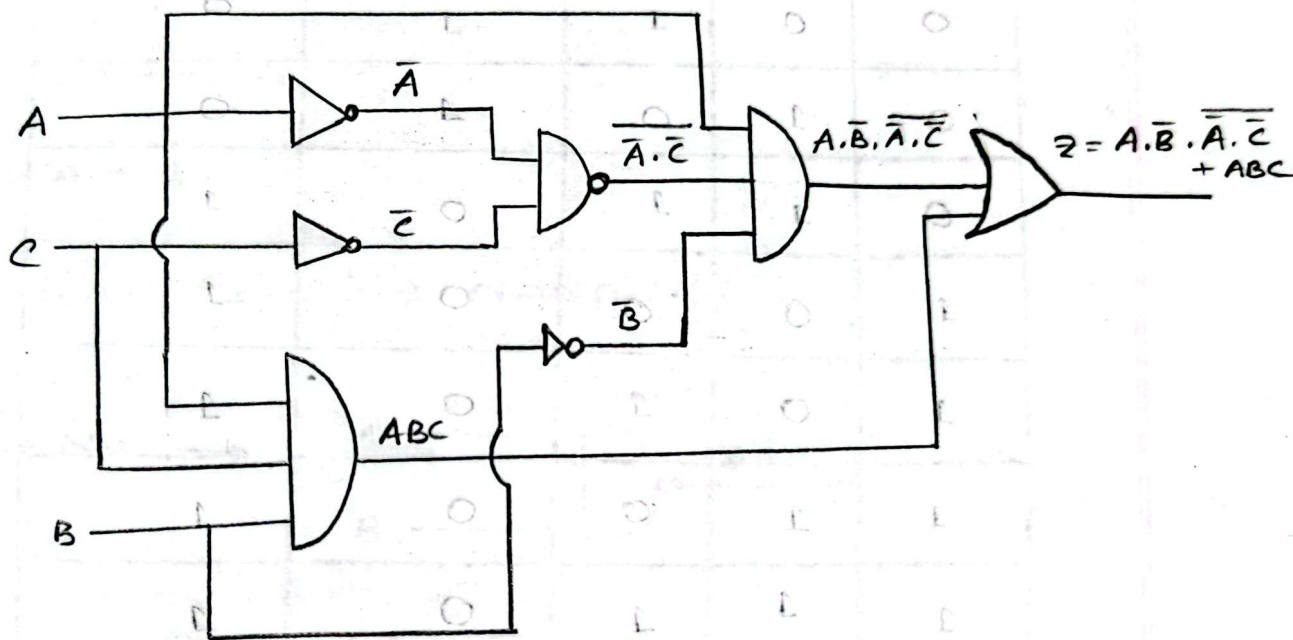
$$= ABC + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}\bar{C}$$

$$\begin{aligned}
 &(111) \quad (101) \quad (100) \quad (110) \quad (001) \\
 &\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 &7 \quad 5 \quad 4 \quad 6 \quad 1
 \end{aligned}$$

$$F(A, B, C) = \sum m (m_1, m_4, m_5, m_6, m_7)$$

[minterm]

• logic gate Simplification



$$\rightarrow A\bar{B} \cdot \bar{A} \cdot \bar{C} + ABC$$

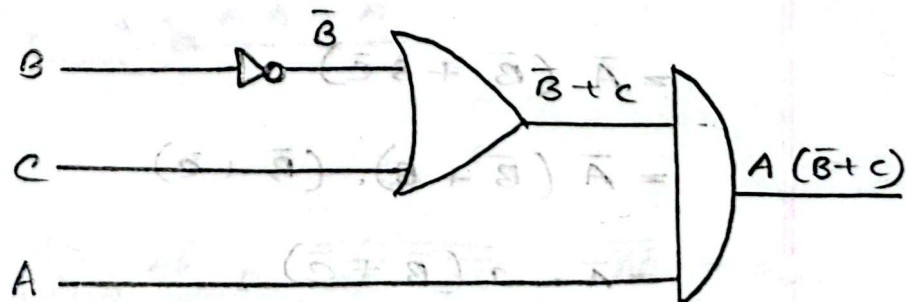
$$= A\bar{B} (\bar{A} + \bar{C}) + ABC$$

$$= A\bar{B} (A + C) + ABC$$

$$= A\bar{B} + AC (B + \bar{B})$$

$$= A\bar{B} + AC$$

$$= A (\bar{B} + C)$$



- Truth table to logic function implementation and Simplification:-

Output

A	B	C	T ₁	T ₂
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

$$\begin{aligned}
 T_1 &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} \\
 &= \bar{A}\bar{B}(C + \bar{C}) + \bar{A}B.\bar{C} \\
 &= \bar{A}\bar{B} + \bar{A}B\bar{C} \\
 &= \bar{A}(\bar{B} + B\bar{C}) \\
 &= \bar{A}(\bar{B} + B)(\bar{B} + \bar{C}) \\
 &= \bar{A}.1(\bar{B} + \bar{C}) \\
 &= \bar{A}\bar{B} + \bar{A}\bar{C}
 \end{aligned}$$

$$T_2 = \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

$$T_2 = (T_1)'$$

$$(T_1)' = \overline{\bar{A} \cdot \bar{B} + \bar{A} \cdot \bar{C}}$$

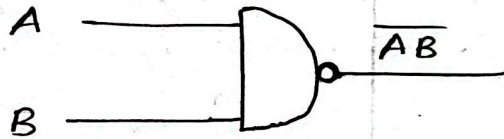
$$= \overline{\bar{A} \cdot \bar{B}} \cdot \overline{\bar{A} \cdot \bar{C}}$$

$$= (A + B)(A + C)$$

$$= A + BC$$

□ Universal Gate $\rightarrow$ NAND, NOR.

• NAND $\rightarrow$



NAND

NOT (1)

AND (2)

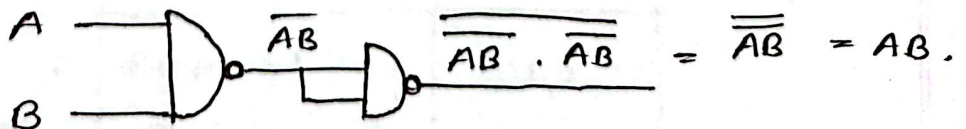
OR (3)

[Gate Needed]

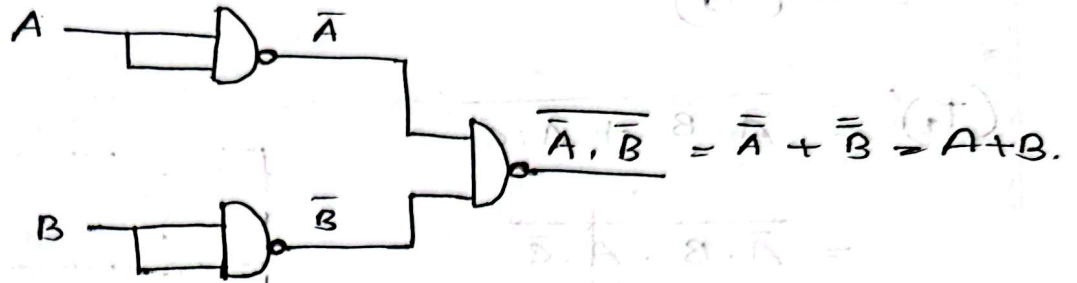
• NOT $\rightarrow$



• AND $\rightarrow$

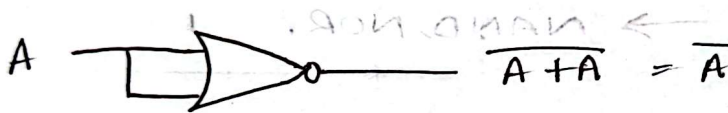


• OR

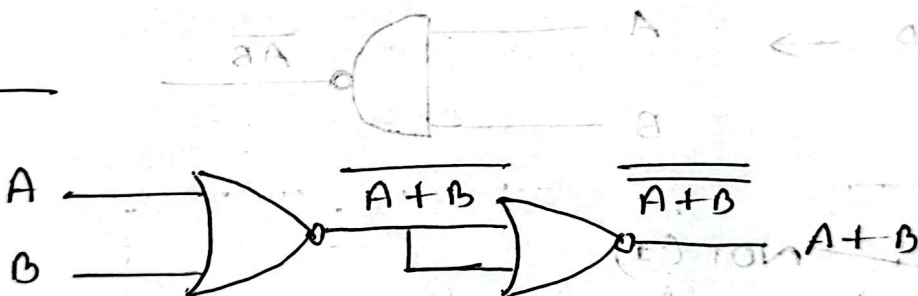


• NAND to NOT & OR gate :

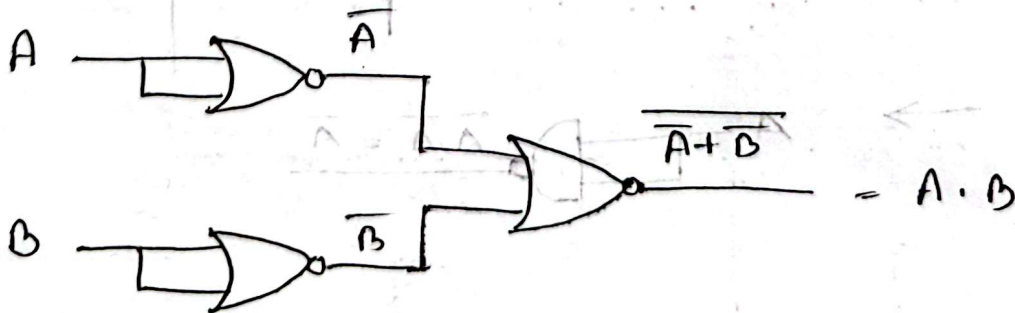
NOT



OR



AND



K-map

• two variable

		0	1
		A	B
0	A	00 m_0	01 m_1
1	A	10 m_2	11 m_3

• three variable

		00	01	11	10
	B/C	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
0	$\bar{A}$	$\bar{A}\bar{B}\bar{C}$ 000 m_0	$\bar{A}\bar{B}C$ 001 m_1	$\bar{A}B\bar{C}$ 011 m_3	$\bar{A}BC$ 010 m_2
1	A	$A\bar{B}\bar{C}$ 100 m_4	$A\bar{B}C$ 101 m_5	$AB\bar{C}$ 111 m_7	ABC 110 m_6

• four variable

		00	01	11	10
	CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
00	$\bar{A}\bar{B}$	0000 m_0	0001 m_1	0011 m_3	0010 m_2
01	$\bar{A}B$	0100 m_4	0101 m_5	0111 m_7	0110 m_6
11	AB	1100 m_{12}	1101 m_{13}	1111 m_{15}	1110 m_{14}
10	$A\bar{B}$	1000 m_8	1001 m_9	1011 m_{11}	1010 m_{10}

• श्रुतिः

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	0	0	1	1
$\bar{C}D$	0	1	1	1
$C\bar{D}$	1	1	0	0
CD	0	1	0	0

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	1	0	0	1
$\bar{C}D$	0	0	0	0
$C\bar{D}$	0	1	1	0
CD	1	0	0	1

$$F(A,B,C) = \sum (0, 1, 2, 3, 4, 6, 7)$$

0	0	0	0
1	0	0	1
1	0	0	1
0	0	0	1

• $F(A,B,C) = \sum (0, 1, 2, 3, 4, 6, 7)$

	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	m_0 1	m_1 1	m_3 1	m_2 1
A 1	m_4 1	m_5 0	m_7 1	m_6 1

$\bar{A} + B + \bar{C}$

• $F(A, B, C, D) = \sum (0, 1, 2, 7, 8, 9, 10, 12, 14)$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 1	m_1 1	m_3 0	m_2 1
$\bar{A}B$	m_4 0	m_5 0	m_7 1	m_6 0
AB	m_{12} 1	m_{13} 0	m_{15} 0	m_{14} 1
$A\bar{B}$	m_8 1	m_9 1	m_{11} 0	m_{10} 1

∴ Simplified function, $F = \bar{B}\bar{C} + A\bar{D} + \bar{B}\bar{D} + \bar{A}BCD$

• $F(A, B, C, D, E) = \sum m(0, 5, 6, 8, 9, 10, 11, 16, 20, 24, 25, 26, 27)$

$A = 0 \Rightarrow \bar{A}$

$A = 1 \Rightarrow A$

SP-25

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	m_0 1	m_1 0	m_3 0	m_2 0
$\bar{B}C$	m_4 0	m_5 1	m_7 0	m_6 1
BC	m_{12} 0	m_{13} 0	m_{15} 0	m_{14} 0
$B\bar{C}$	m_8 1	m_9 1	m_{11} 1	m_{10} 1

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	m_{16} 1	m_{17} 0	m_{19} 0	m_{18} 0
$\bar{B}C$	m_{20} 1	m_{21} 0	m_{23} 0	m_{22} 0
BC	m_{28} 0	m_{29} 0	m_{31} 0	m_{30} 0
$B\bar{C}$	m_{24} 1	m_{25} 1	m_{27} 1	m_{26} 1

$= B\bar{C} + \bar{C}\bar{D}\bar{E} + A\bar{B}\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}E + \bar{A}\bar{B}CDE$

• $F(A, B, C, D) = \sum (1, 3, 7, 11, 15)$

$d(A, B, C, D) = \sum (0, 2, 5)$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 X	m_1 1	m_3 1	m_2 X
$A\bar{B}$	m_4 0	m_5 X	m_7 1	m_6 0
AB	m_{12} 0	m_{13} 0	m_{15} 1	m_{14} 0
$\bar{A}B$	m_8 0	m_9 0	m_{11} 1	m_{10} 0

$$\bar{A}D + CD$$

$$= D(\bar{A} + C)$$

$\bar{A}\bar{B} + CD$ ←

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 X	m_1 1	m_3 1	m_2 1
$A\bar{B}$	m_4 0	m_5 X	m_7 1	m_6 0
AB	m_{12} 0	m_{13} 0	m_{15} 1	m_{14} 0
$\bar{A}B$	m_8 0	m_9 0	m_{11} 1	m_{10} 0

POS

• Simplification using K-map

$$* F(A, B, C, D) = \pi (0, 1, 2, 3, 4, 10, 11)$$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 0	m_1 0	m_3 0	m_2 0
$\bar{A}B$	m_4 0	m_5 1	m_7 1	m_6 1
AB	m_{12} 1	m_{13} 1	m_{15} 1	m_{14} 1
$A\bar{B}$	m_8 1	m_9 1	m_{11} 0	m_{10} 0

{ pos - 0 के सूचित करें }

$$F = \bar{A}\bar{B} + \bar{A}\bar{C}\bar{D} + \bar{B}C$$

$$F = (F')'$$

$$F' = \overline{\bar{A}\bar{B} + \bar{A}\bar{C}\bar{D} + \bar{B}C}$$

$$F' = \overline{\bar{A}\bar{B}} \cdot \overline{\bar{A}\bar{C}\bar{D}} \cdot \overline{\bar{B}C}$$

$$= (A+B) (A+C+D) (B+\bar{C})$$

Grouping

4 → common 2

2 → 3

8 →

[0 मिले या function पाया उसे $F'(POS)$]

• Codes

BCD $\Rightarrow$ 15 $\Rightarrow$ $\overbrace{0001}^1 \overbrace{0101}^5 \rightarrow$ BCD

BCD + 0011

	BCD	Excess-3	2421
0	0000	0011	0000
1	0001	0100	0001
2	0010	0101	0010
3	0011	0110	0011
4	0100	0111	0100
5	0101	1000	0101 1011
6	0110	1001	0110 1100
7	0111	1010	1101
8	1000	1011	1110
9	1001	1100	1111

	BCD				Excess-3			
	A	B	C	D	x	y	z	w
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

$x =$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 0	m_1 0	m_3 0	m_2 0
$\bar{A}B$	m_4 0	m_5 1	m_7 1	m_6 1
AB	m_{12} X	m_{13} X	m_{15} X	m_{14} X
$A\bar{B}$	m_8 1	m_9 1	m_{11} X	m_{10} X

{যে নম্বরগুলো নাহে
যেগুলোতে 1 don't
care হবে অর্থাৎ
X হবে.}

$$x = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$$

$$\hookrightarrow A + BD + BC$$

$y =$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 0	m_1 1	m_3 1	m_2 1
$\bar{A}B$	m_4 1	m_5 0	m_7 0	m_6 0
AB	m_{12} X	m_{13} X	m_{15} X	m_{14} X
$A\bar{B}$	m_8 0	m_9 1	m_{11} X	m_{10} X

$$y = \bar{B}D + \bar{B}C + B\bar{C}\bar{D}$$

$z =$

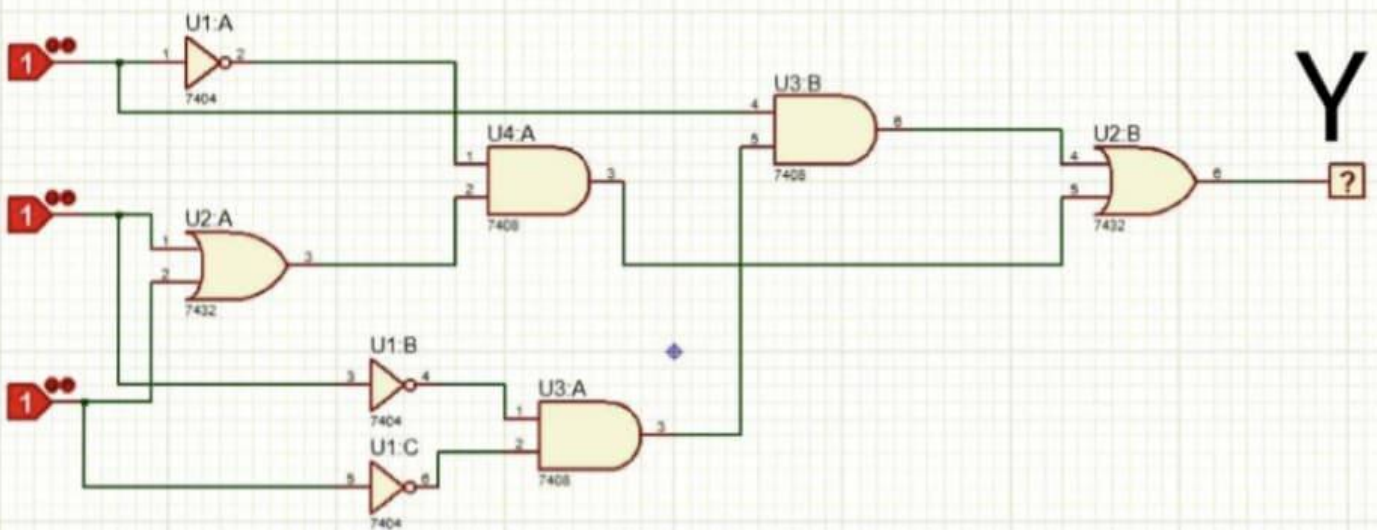
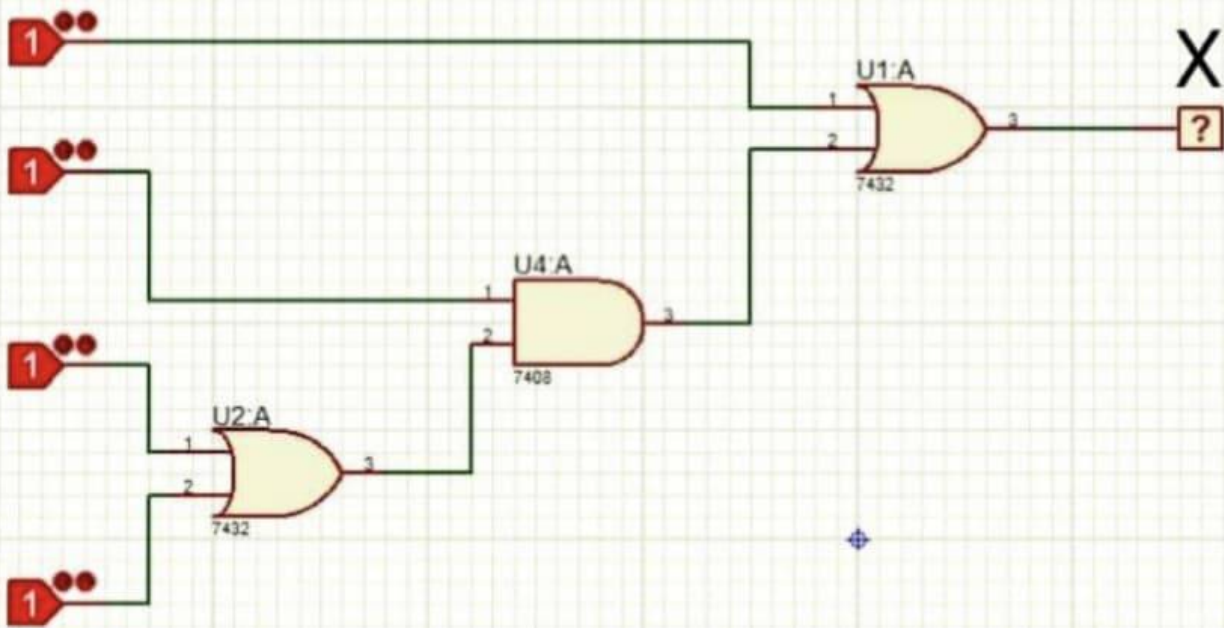
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 1	m_1 0	m_3 1	m_2 0
$\bar{A}B$	m_4 1	m_5 0	m_7 1	m_6 0
AB	m_{12} X	m_{13} X	m_{15} X	m_{14} X
$A\bar{B}$	m_8 1	m_9 0	m_{11} X	m_{10} X

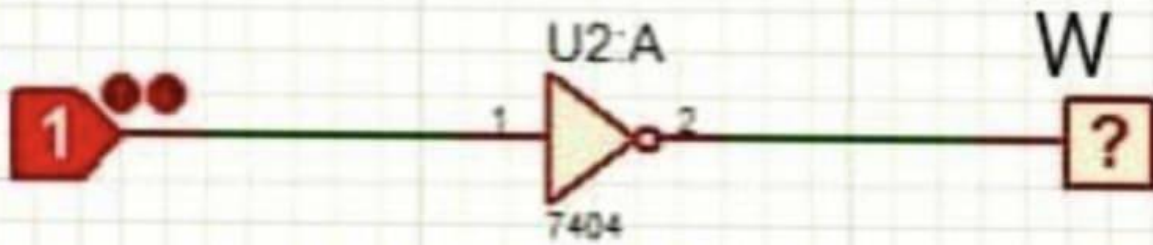
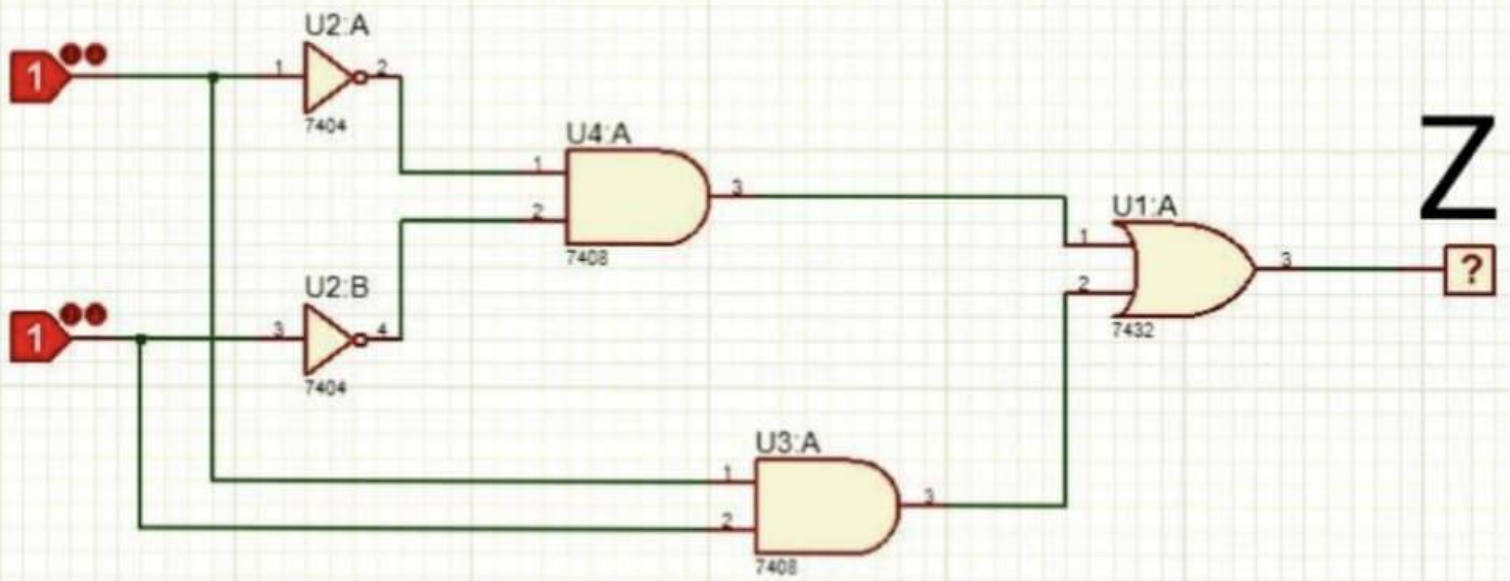
$$z = \bar{C}\bar{D} + CD$$

$w =$

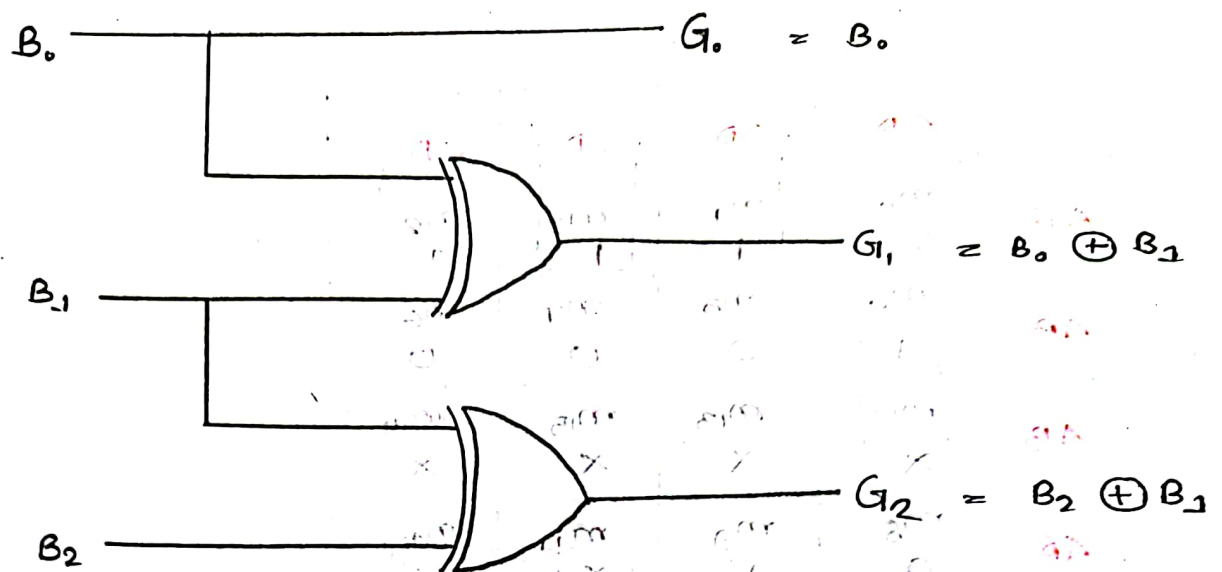
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0 1	m_1 0	m_3 0	m_2 1
$\bar{A}B$	m_4 1	m_5 0	m_7 0	m_6 1
AB	m_{12} X	m_{13} X	m_{15} X	m_{14} X
$A\bar{B}$	m_8 1	m_9 0	m_{11} X	m_{10} X

$$w = \bar{D}$$

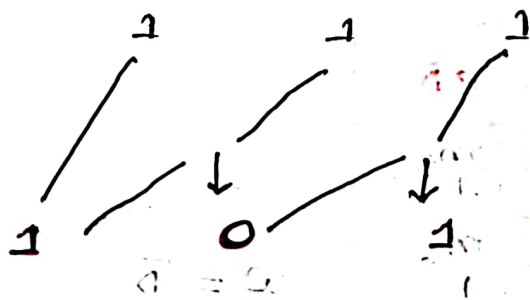
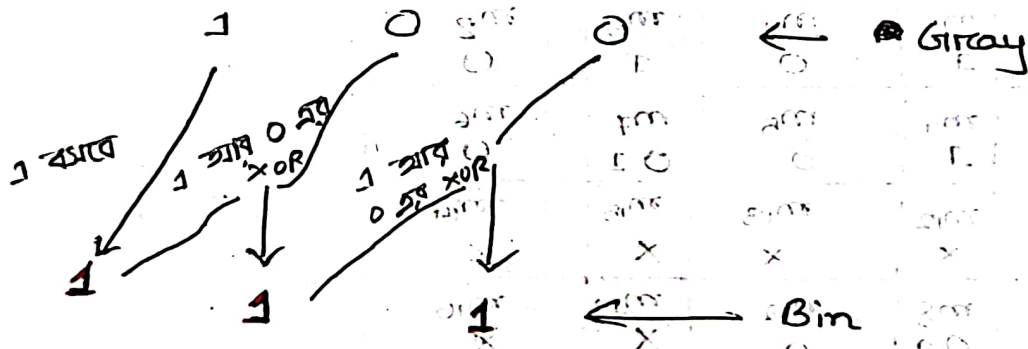




• Bin to Gray



• Gray to Bin



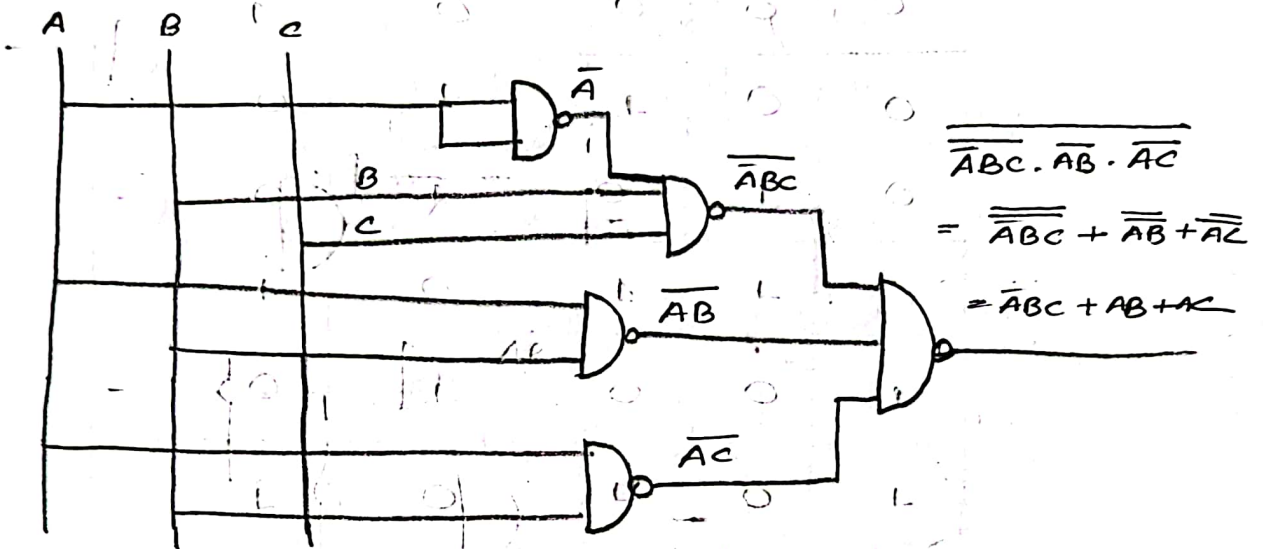
• Parity Check

x	y	z	E	O
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

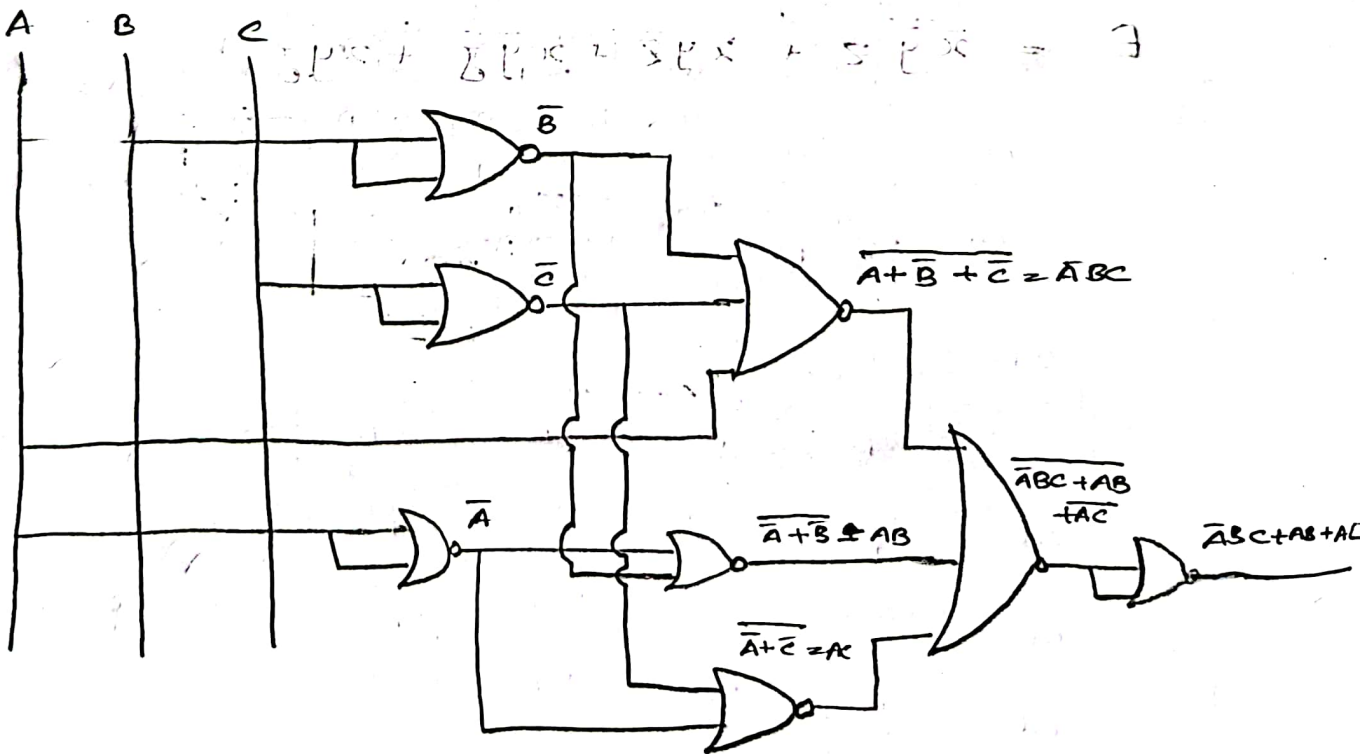
$$E = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

• NAND, NOR Implementation:

Q * $\bar{A}BC + AB + AC \rightarrow$ NAND Gate implementation

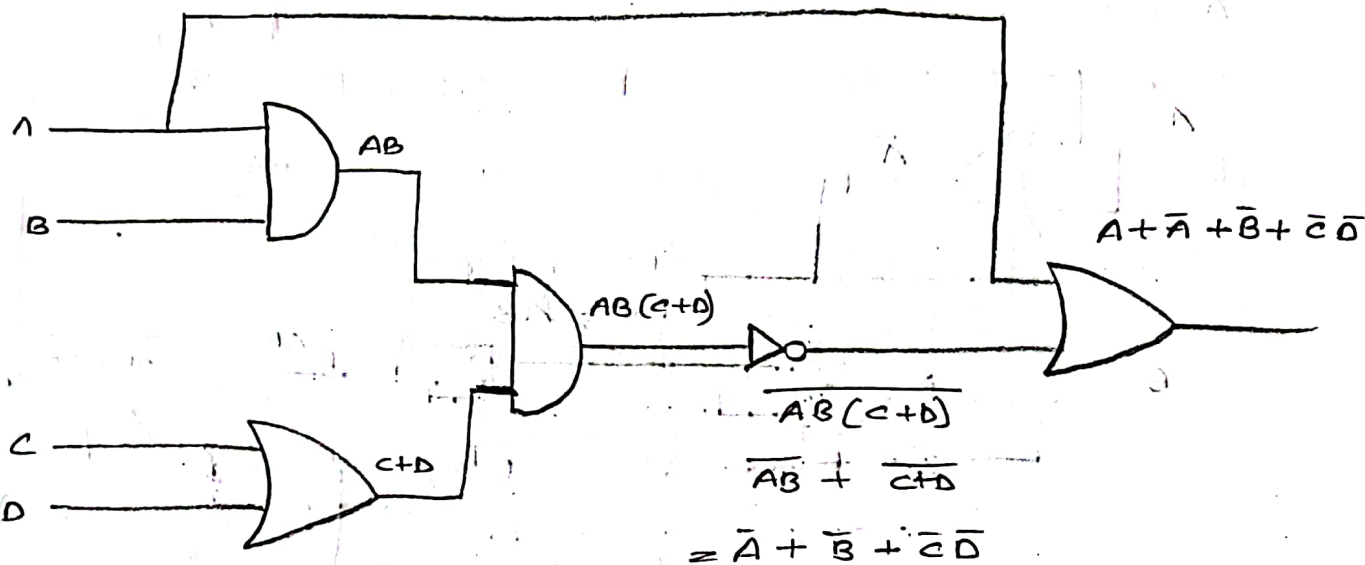


Q * $\bar{A}BC + AB + AC \rightarrow$ NOR Gate implementation

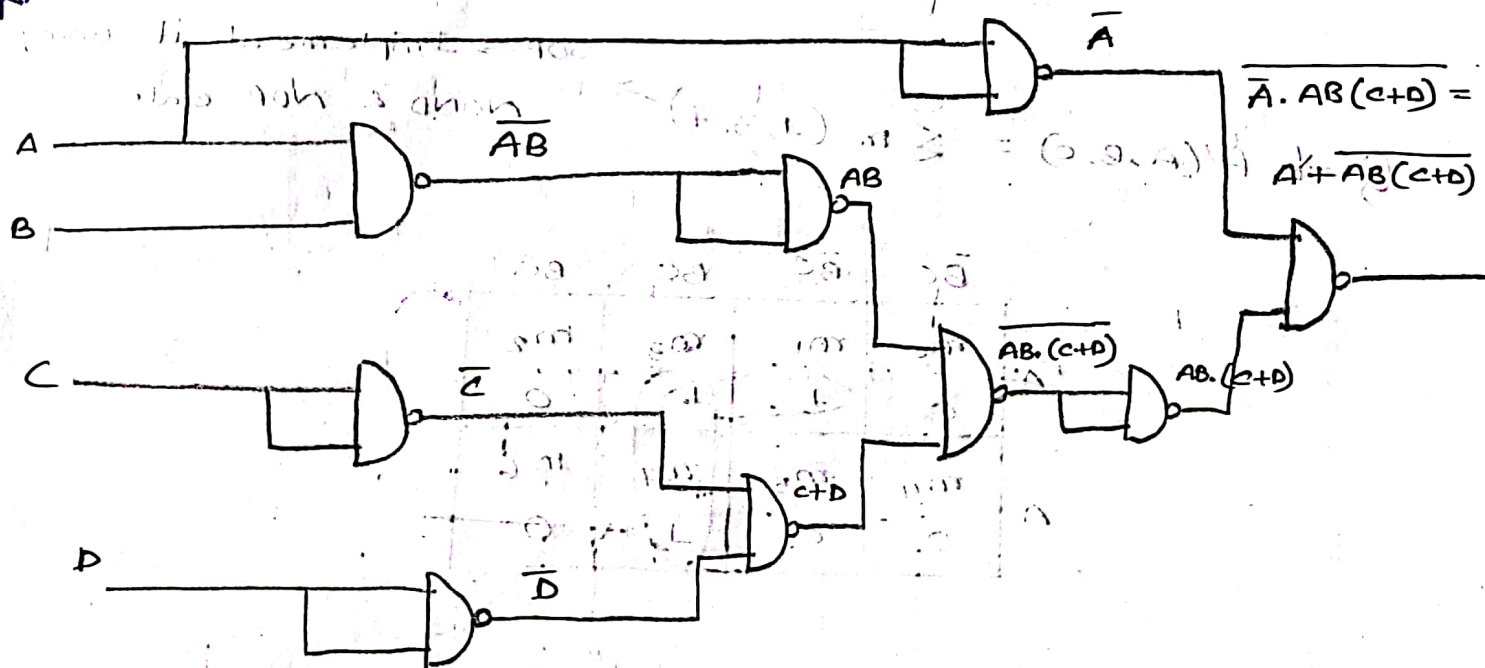


9 *

NAND
 AND (2)
 NOT (1)
 OR (3)



NAND

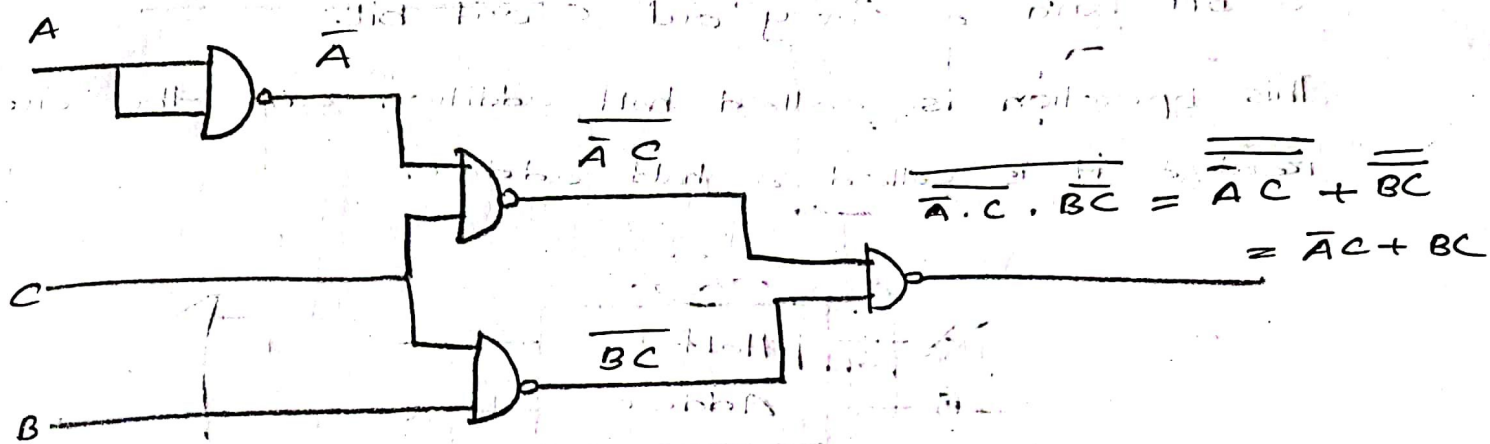


NOR

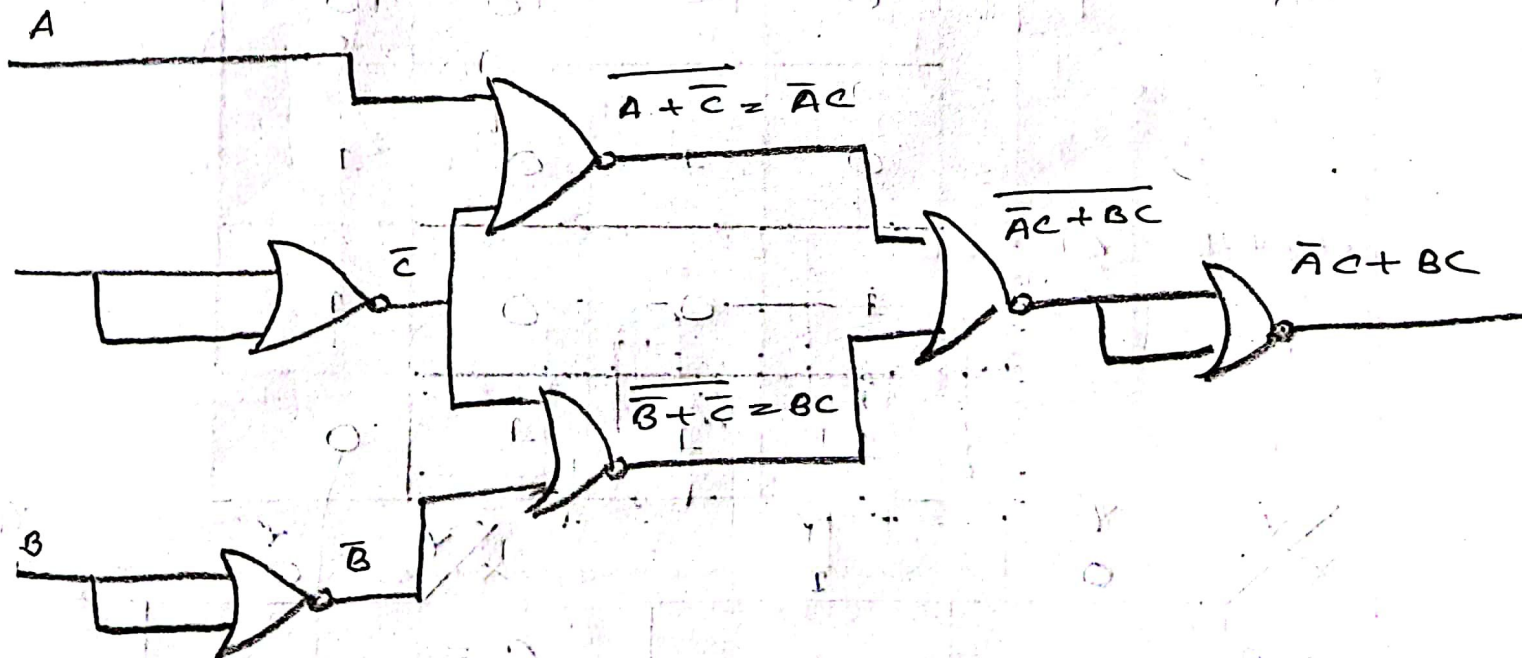

$$Q * F(A, B, C) = \sum_m (1, 3, 7) \rightarrow$$
$$f = \bar{A}C + BC$$

$$Z = C(\bar{A} + B)$$

NAND →



NOR

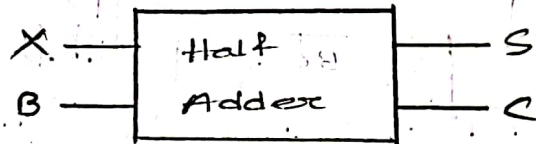


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• Half Adder :

Adding two single-bit binary values, A, B produces a sum S bit and a carry out C-out bit.

This operation is called half addition and the circuit to realize it is called a half adder.



• Truth Table

Inputs		Outputs	
X	Y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

X \ Y	0	1
0	0	1
1	1	0

$$S = X\bar{Y} + \bar{X}Y$$

$$S = X \oplus Y$$

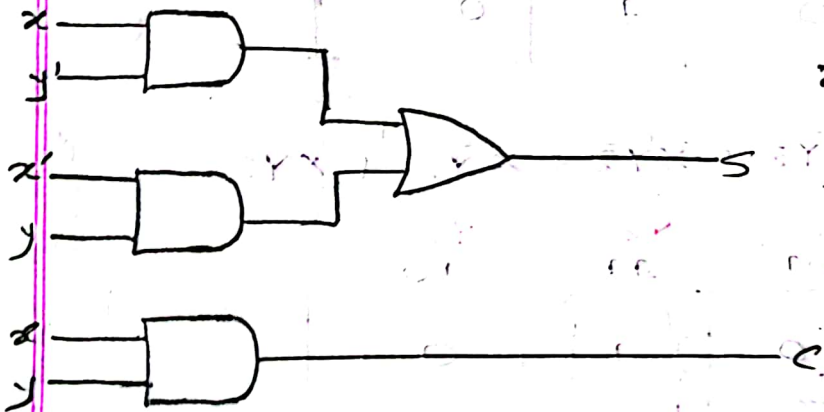
X \ Y	0	1
0	0	0
1	0	1

$$C = XY$$

Implementation of Half Adders

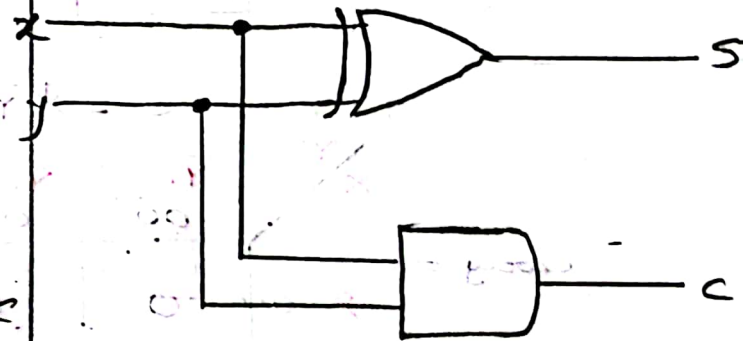
(a) $S = xy + \bar{x}\bar{y}$

$C = xy$



(b) $S = x \oplus y$

$C = xy$



• Full Adder :

A combinational circuit that adds 3 input bits to generate a sum bit and a carry bit.

• Truth Table

Inputs			Outputs	
X	Y	Z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Sum = (S)

$x \backslash yz$	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
0	0	1	0	1
1	1	0	1	0

$$S = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy z$$

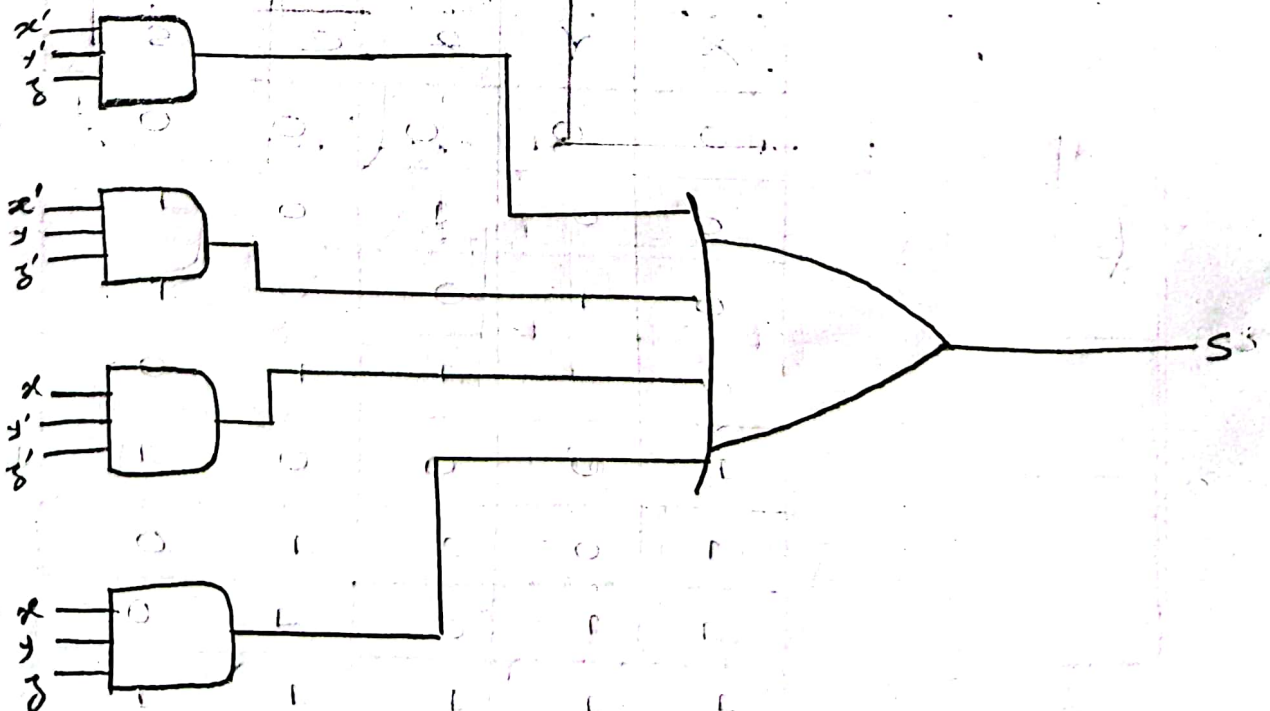
Carry =

$x \backslash yz$	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
0	0	0	1	0
1	0	1	1	1

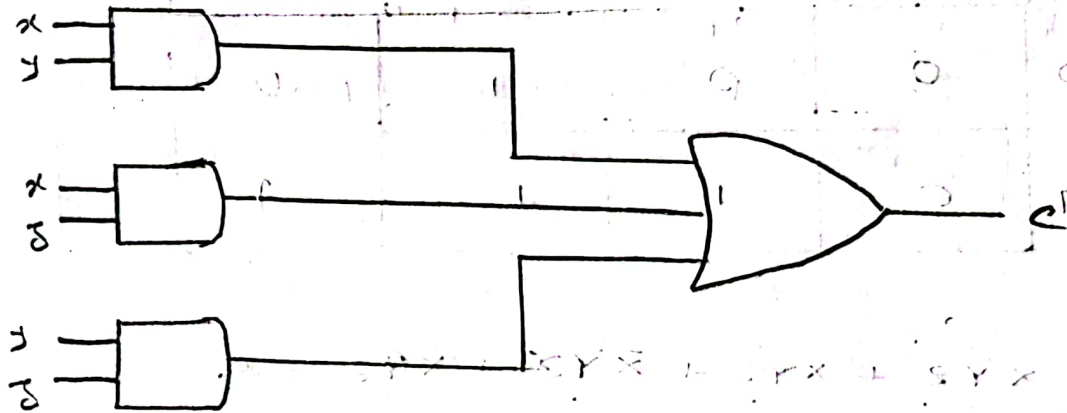
$$C = xy + xz + yz$$

Implementation of full Adder in SOP

$$S = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy z$$



$$C = xy + \bar{x}y\bar{z} + x\bar{y}z$$



Implementation of full adder with two half-adders and an or-gate: - $(x \oplus y)z + xy$

• Manipulating the Equations :

Sum	$\bar{x}$	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	$y\bar{z}$ 11	yz 10
$\bar{x}$	0	0	1	0	1
x	1	1	0	1	0

$$S = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

$$= \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

$$= \bar{z}(\bar{x}\bar{y} + \bar{x}y + x\bar{y} + xy)$$

$$= \bar{z}(x \oplus y) + \bar{z}(x \oplus y)$$

$$= \bar{z} \oplus (x \oplus y)$$

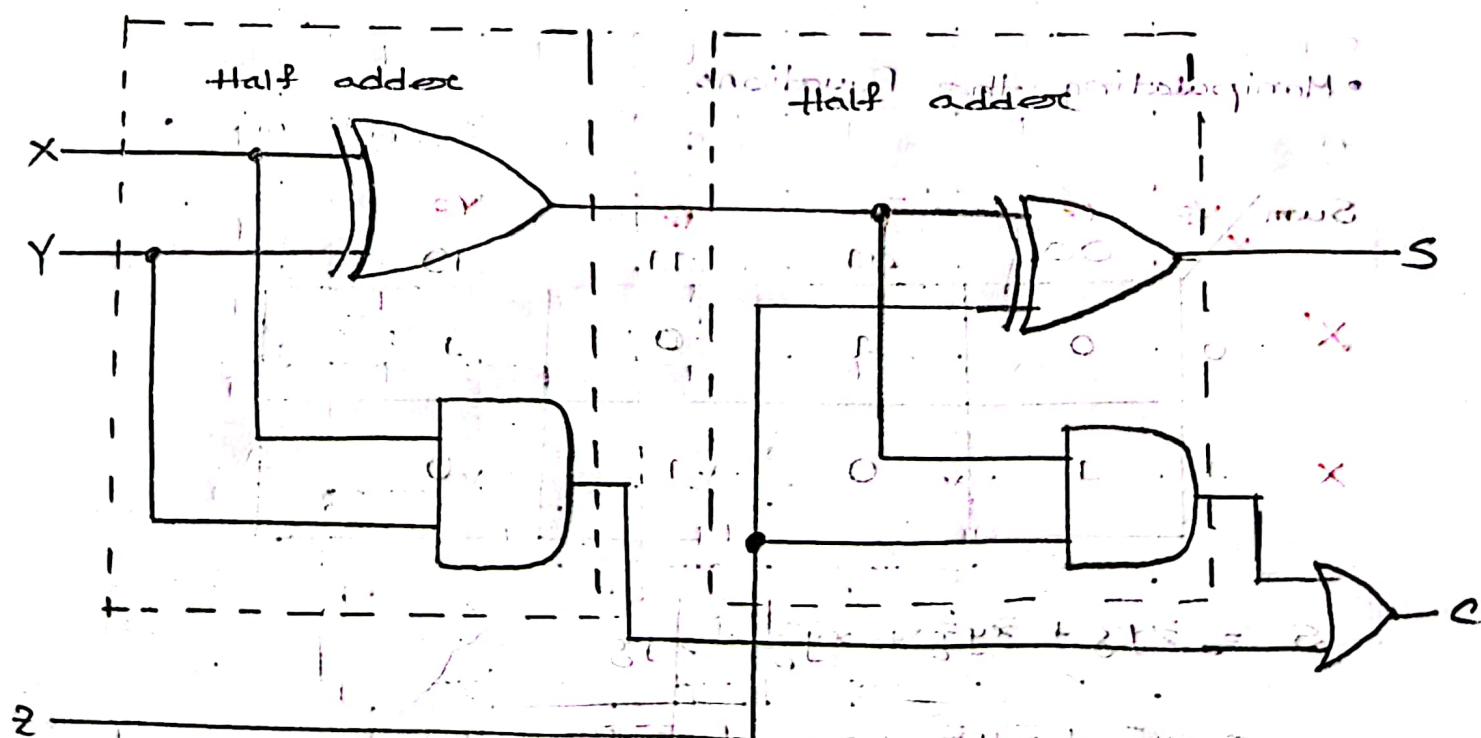
Carry

	$x \bar{y} z$	$\bar{x} \bar{y} \bar{z}$	$\bar{x} y \bar{z}$	$x y \bar{z}$	$x y z$
	00	01	11	10	
x'	0	0	0	1	0
x	1	0	1	1	1

$$C = x \bar{y} z + x y \bar{z} + \bar{x} y z + x y z$$

$$= z(x \bar{y} + \bar{x} y) + x y(z + \bar{z})$$

$$= x y + z(x \oplus y)$$



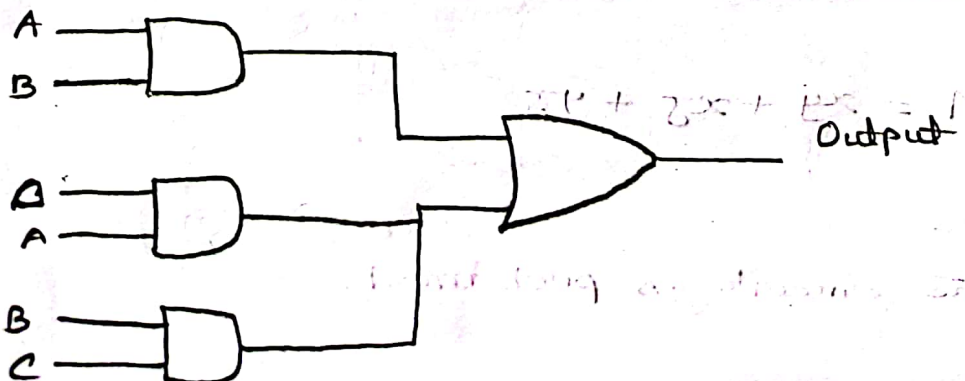
• Think of z as a carry in

Problem : 1. A committee of three individuals decide issues for an organization. Each individual votes either yes or no for each proposal that arises. A proposal is passed if it receives at least two yes votes. Design a circuit that determines whether a proposal passes.

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

	$\bar{B}C$	$\bar{B}\bar{C}$	BC	$B\bar{C}$
$\bar{A}$	0 m_0	0 m_1	1 m_3	0 m_2
A	0 m_4	1 m_5	1 m_7	1 m_6

$$F = BC + AC + AB$$



Problem: 3. Design a combinational circuit with three inputs, x , y and z , and three outputs A , B and C . When the binary input is 0, 1, 2 or 3, the binary output is one greater than the input. When the input is 4, 5, 6 or 7, the binary output is less than the input.

x	y	z	A	B	C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

$A = x y z$	$\bar{y} \bar{z}$ 00	$\bar{y} z$ 01	$y \bar{z}$ 11	$y z$ 10
$\bar{x}$ 0		1	1	1
x 1	1		1	1

$$A = xy + xz + yz$$

Same circuit as problem-1.

$B =$

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	$y\bar{z}$
x	00	01	11	10
$\bar{x}$ 0		1		1
x 1	1		1	

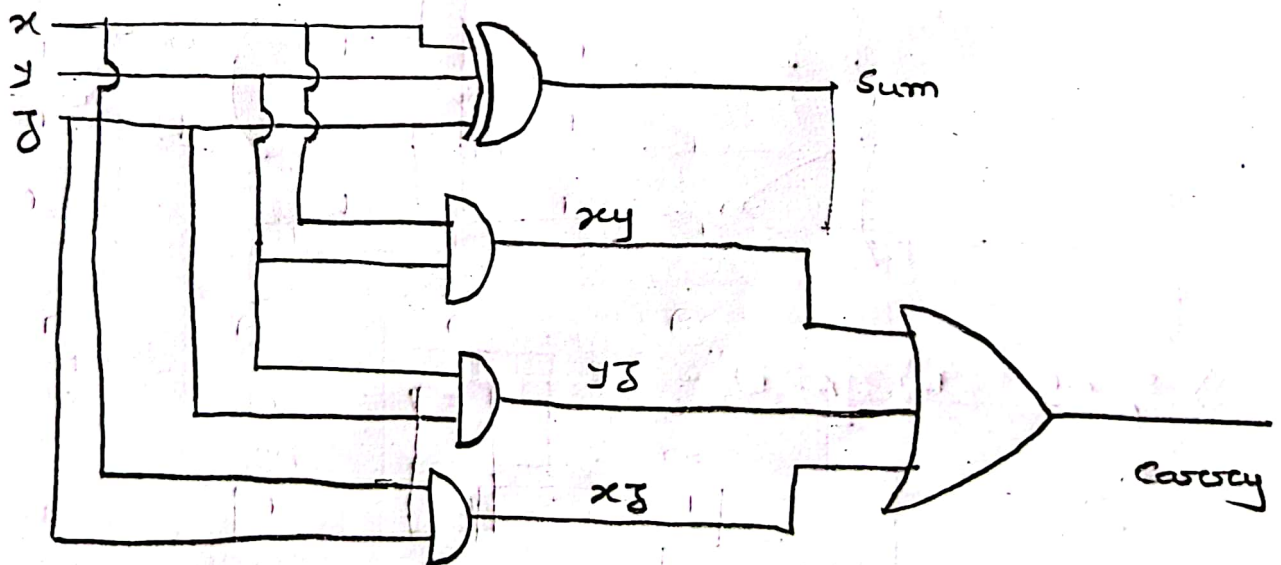
$$B = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

$$= x\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z}$$

$$= \bar{z}(x\bar{y} + \bar{x}y) + z(xy + \bar{x}\bar{y})$$

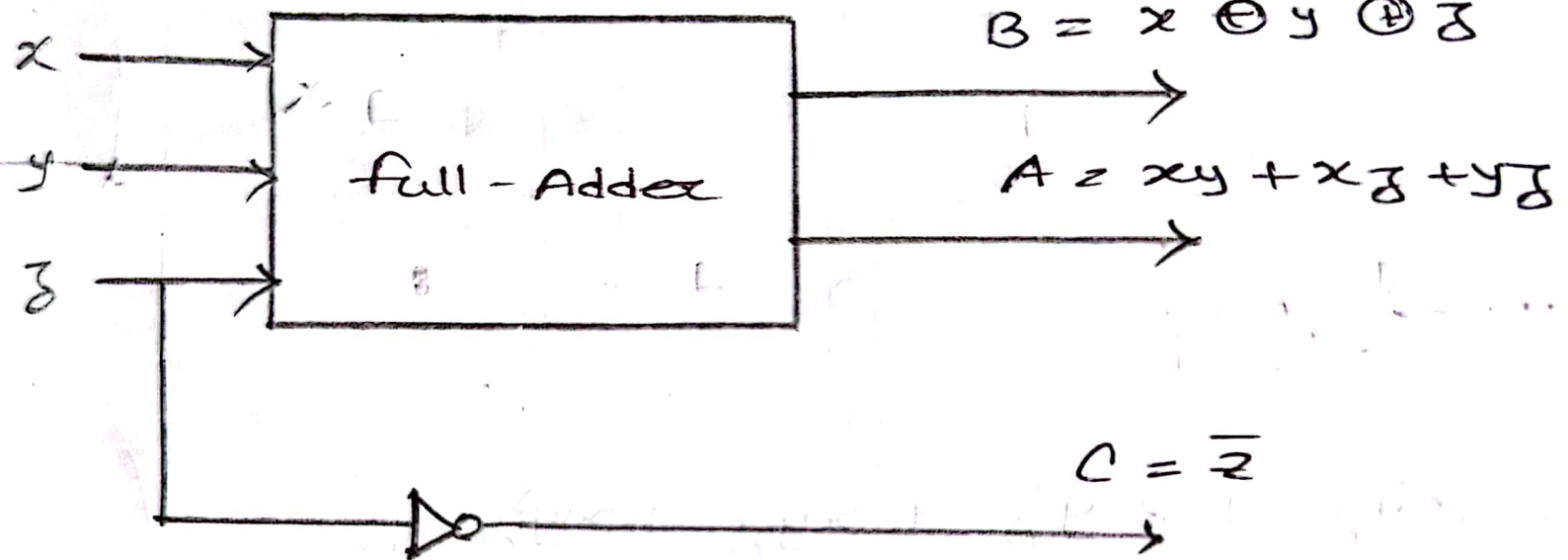
$$= \bar{z}(x \oplus y) + z(\overline{x \oplus y})$$

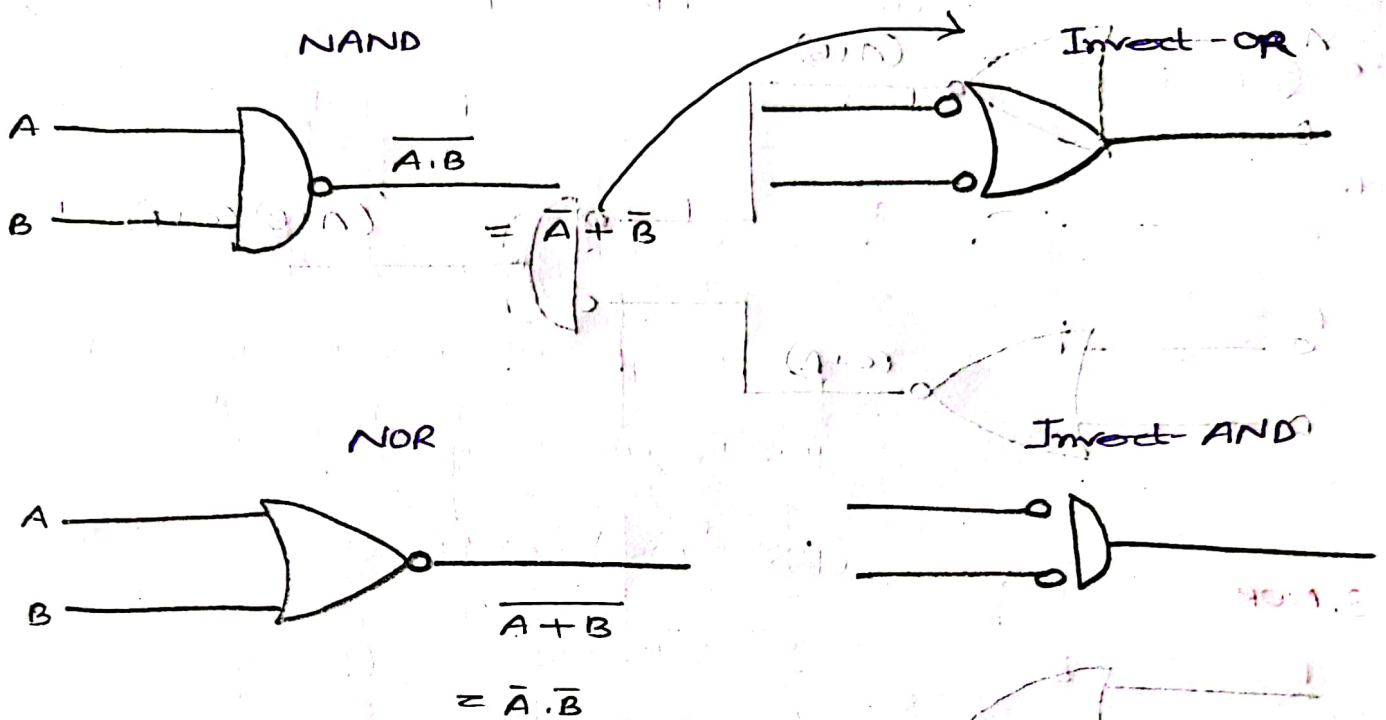
$$= z \oplus (x \oplus y)$$



$$C = \bar{z}$$

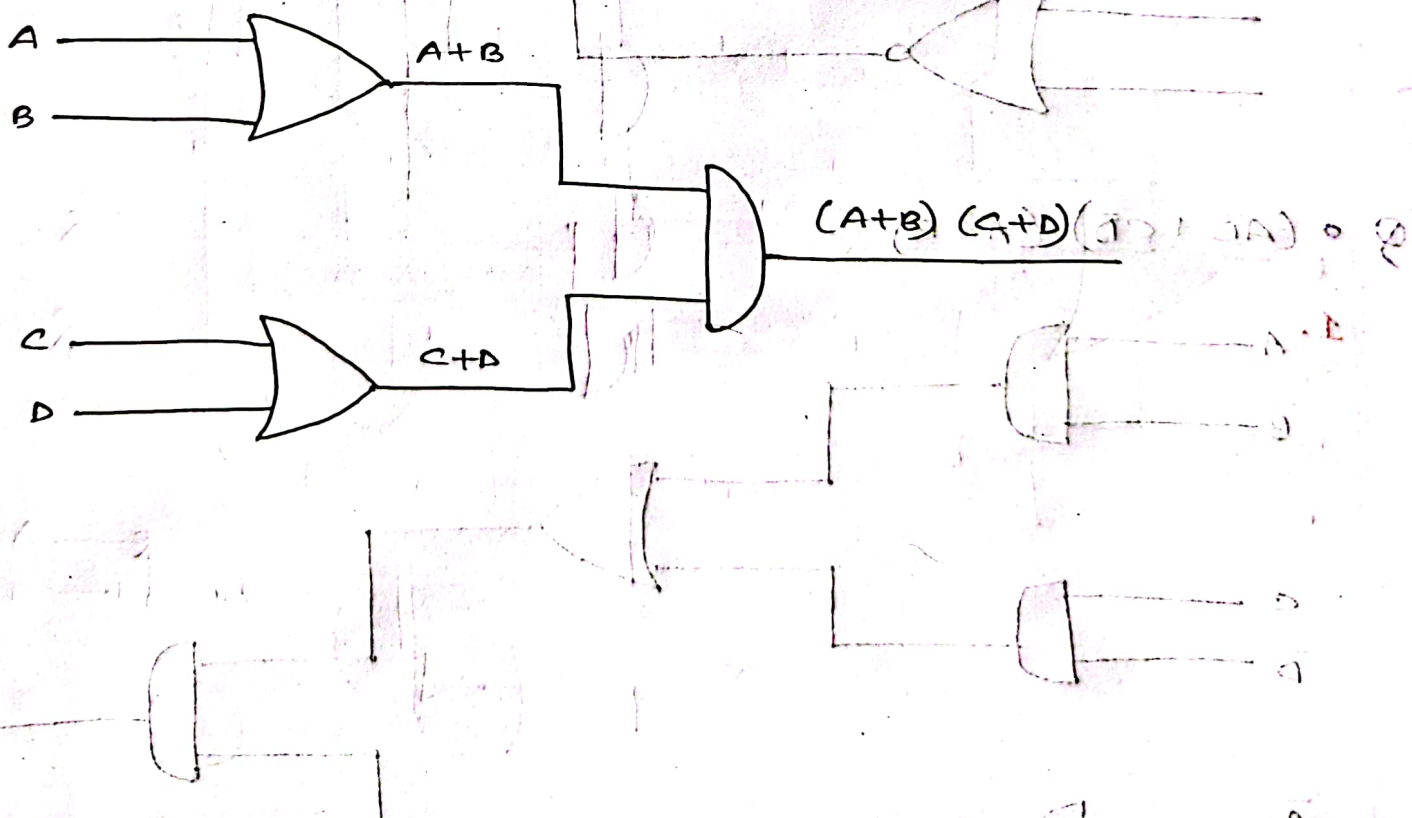




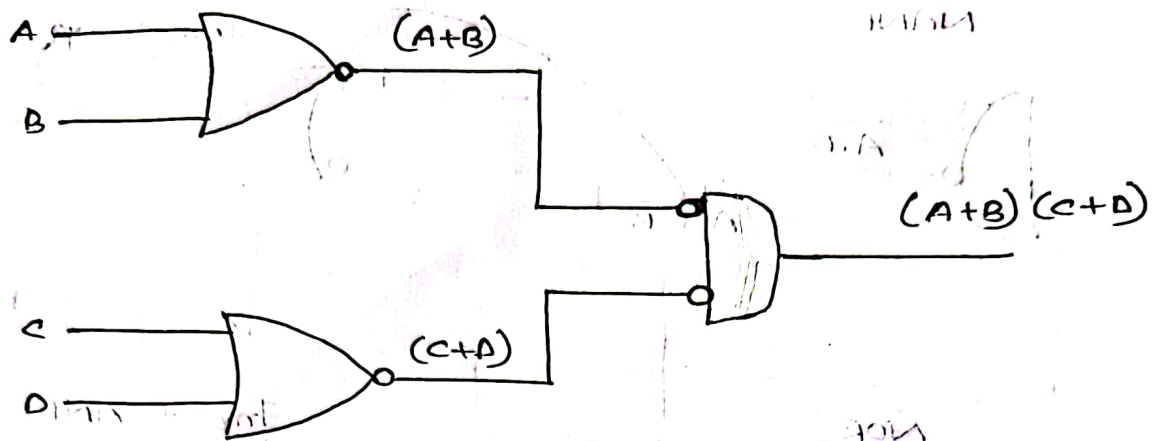


Q • $(A+B)(C+D)$

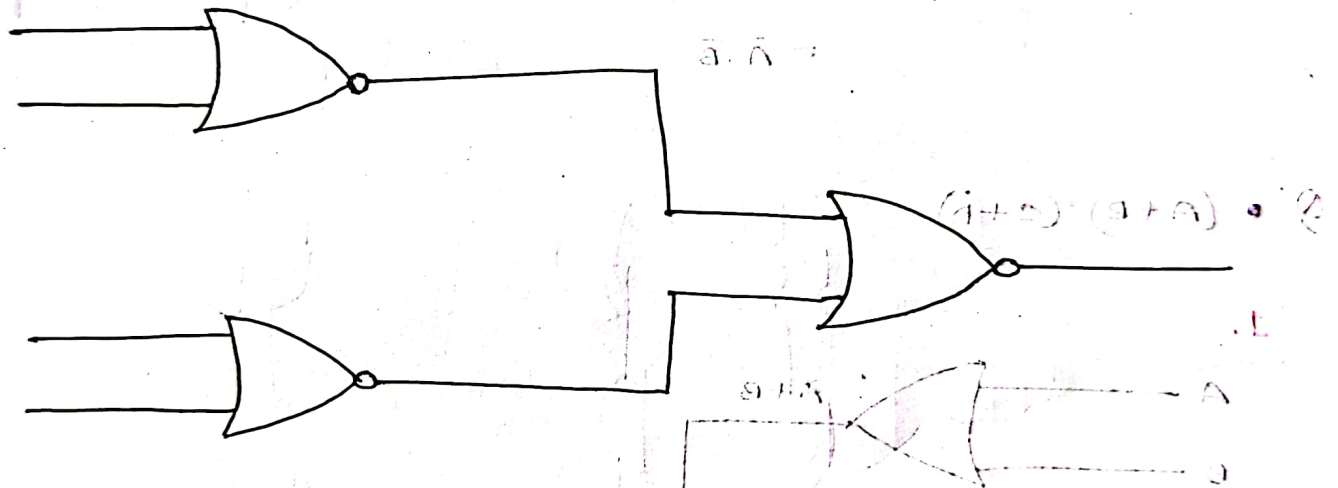
1.



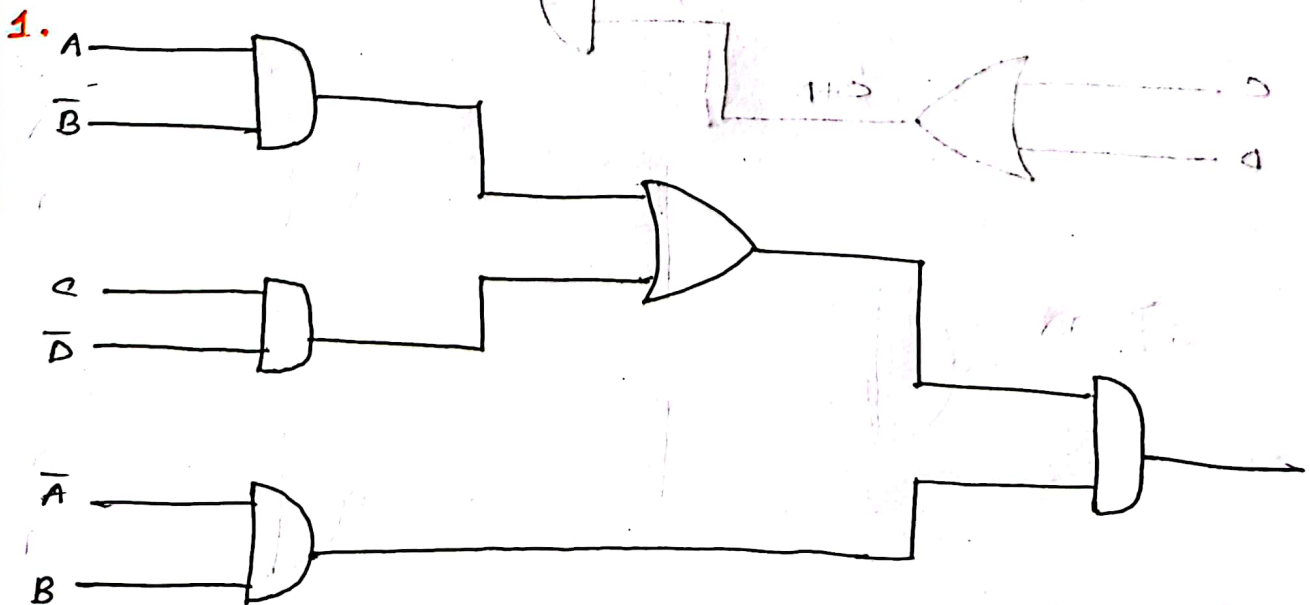
2. NAND



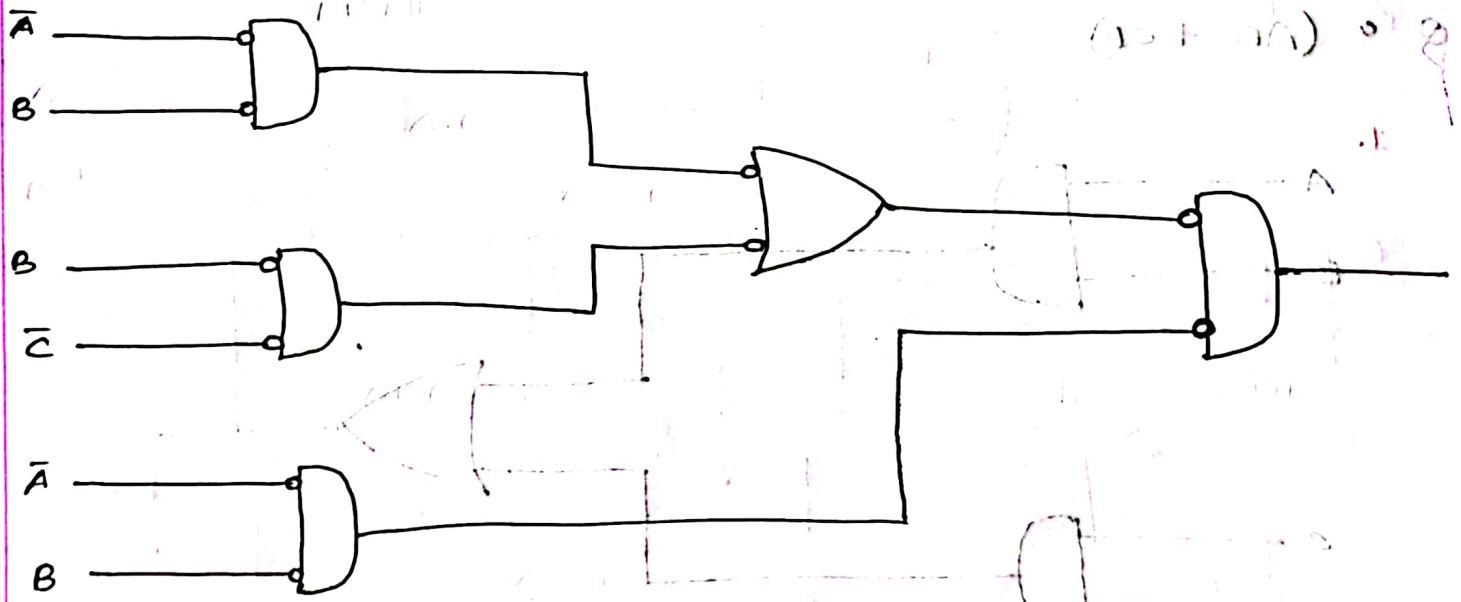
3. NOR



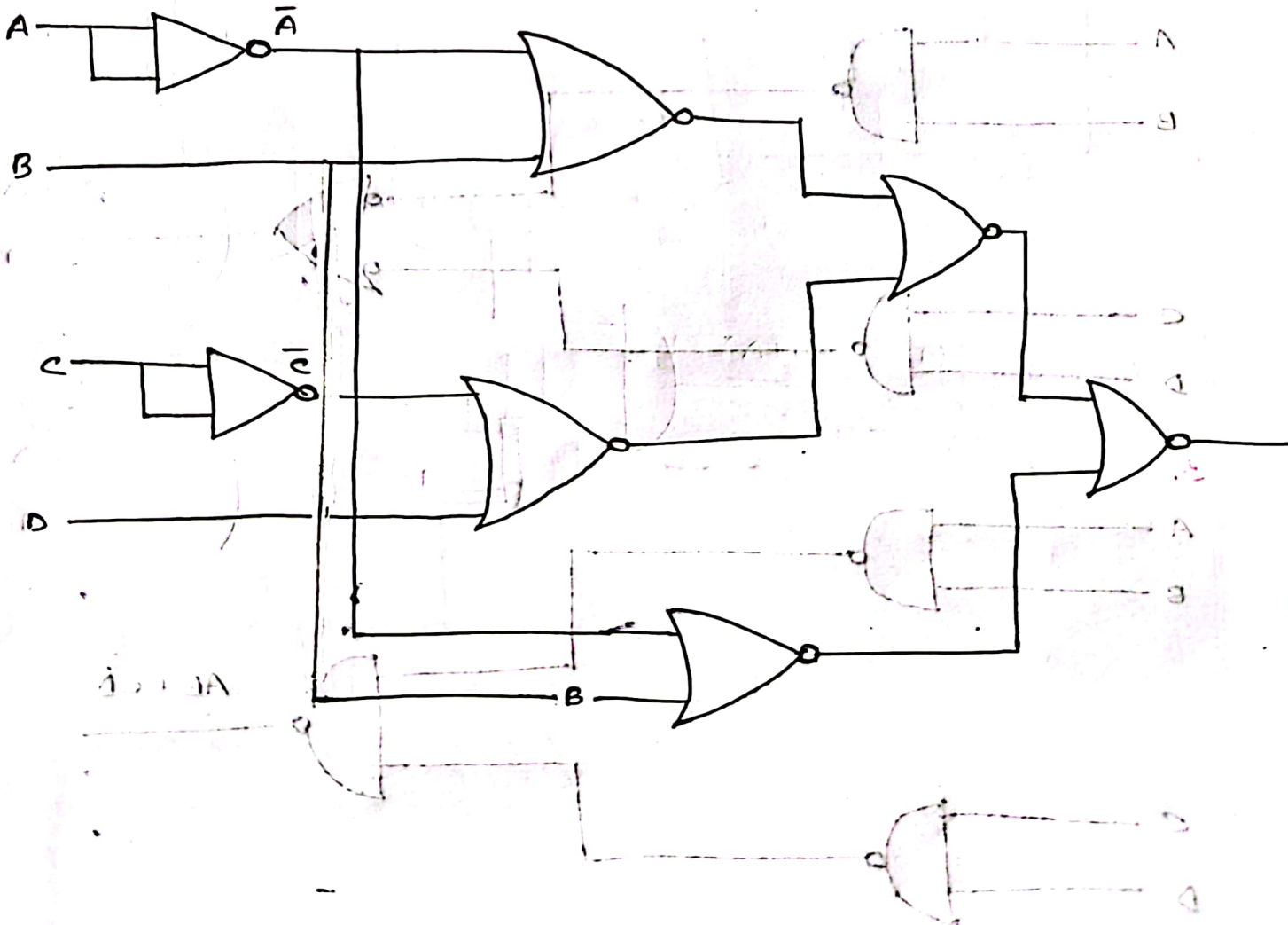
Q • $(\bar{A}\bar{B} + \bar{C}\bar{D})(\bar{A} + B)$



2.



3.

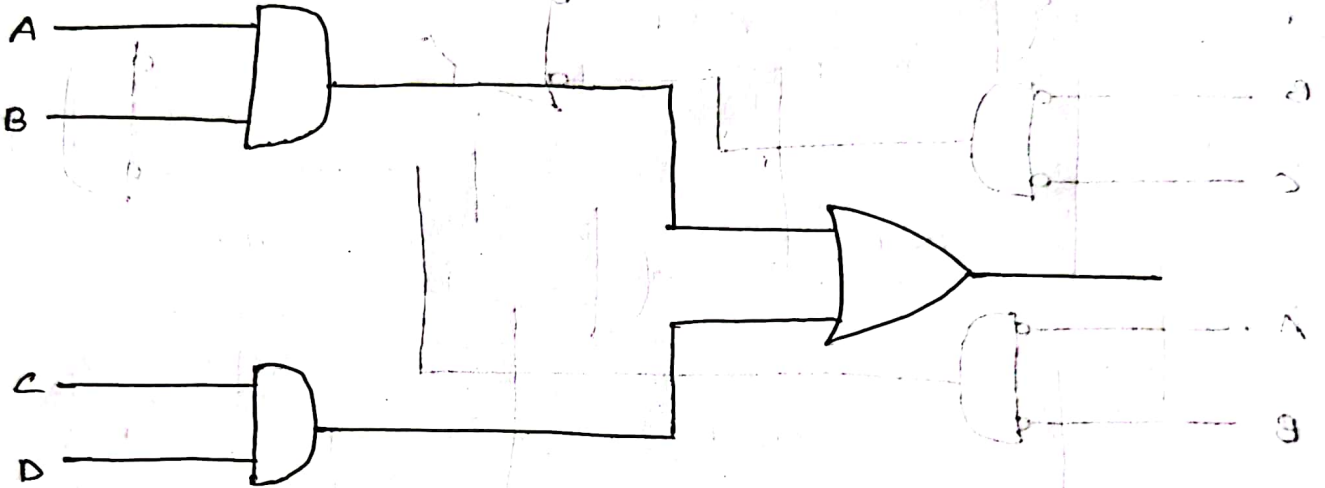


[২ টি বায় শনে কাজ করা হবে]

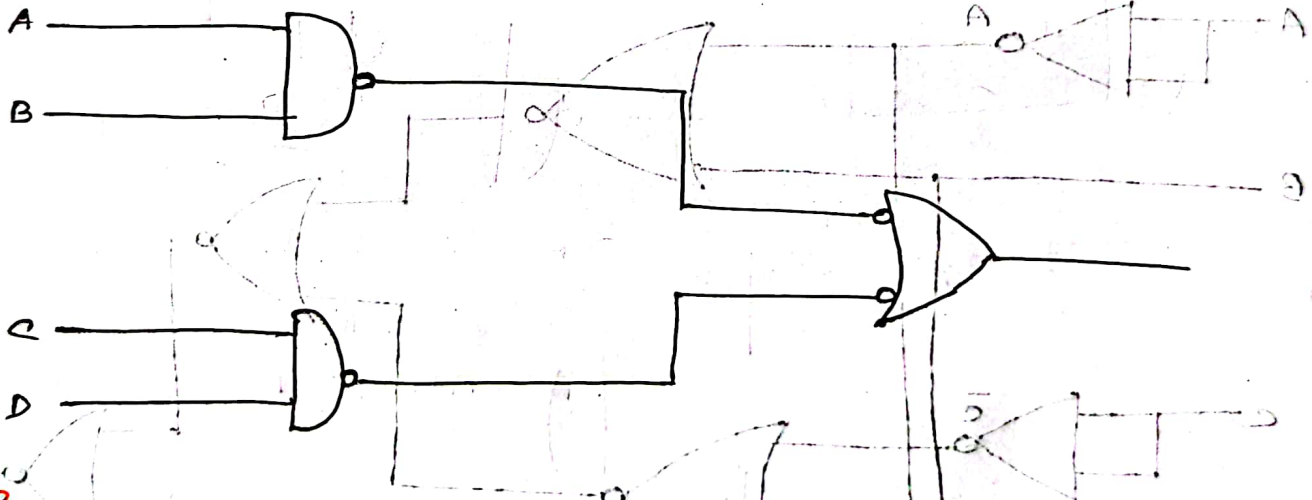
• NAND Implementation :

৯ • $(AB + CD)$

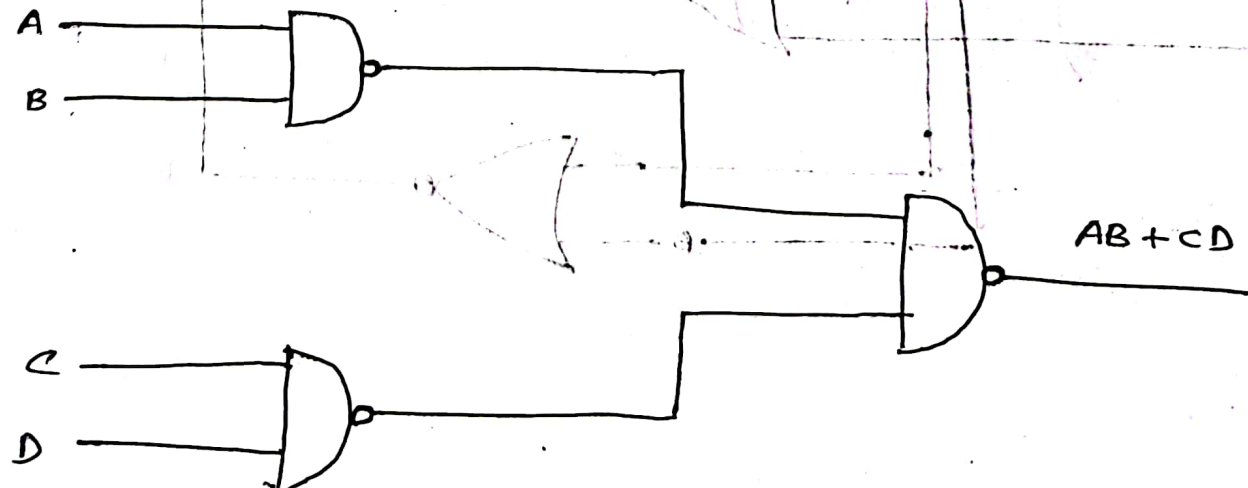
1.



2.

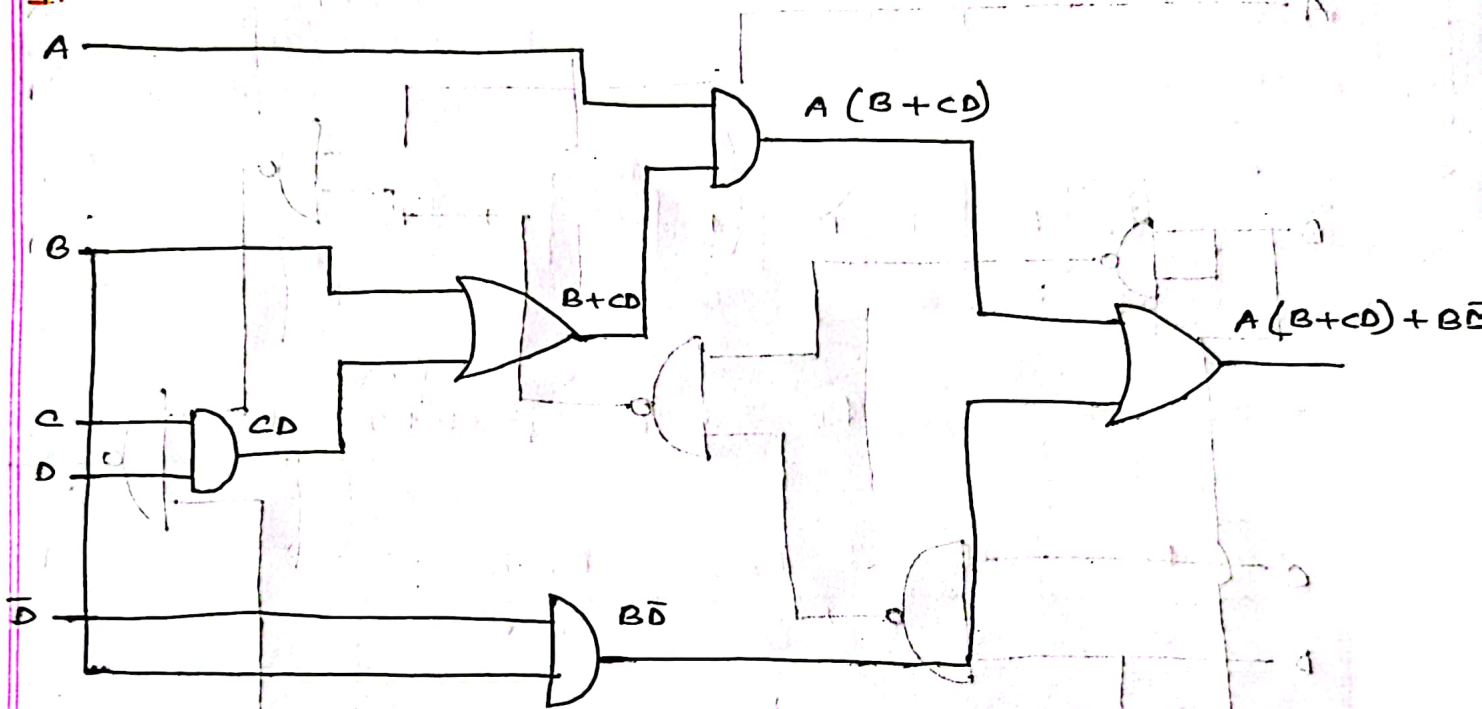


3.

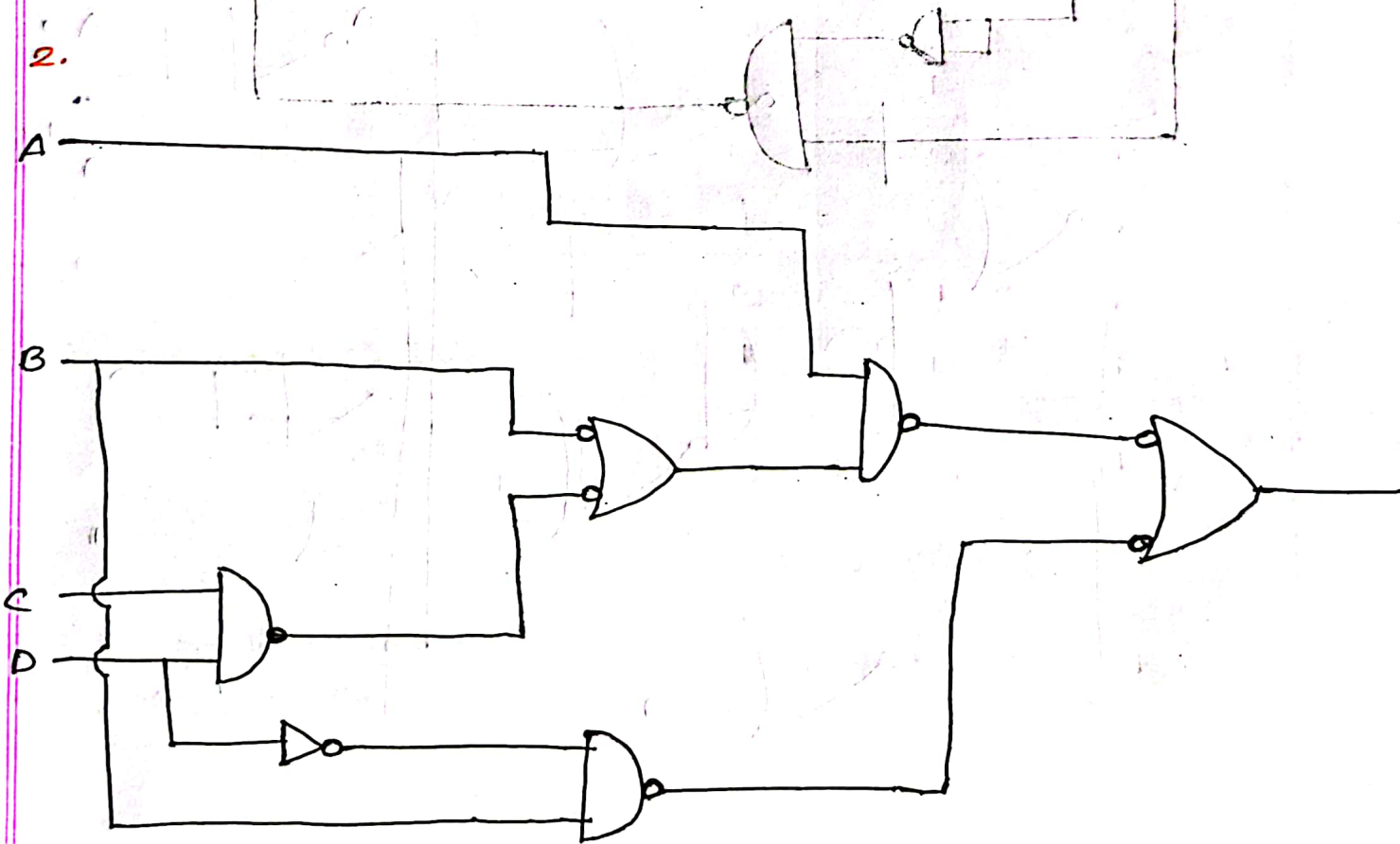


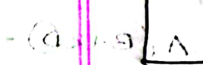
Q. $F = A(B + CD) + B\bar{D}$

1.



2.



$$100 + (100 + 20) \cdot 0.1 = 121 \%$$


• Hamming Code :

Even - Parity

Odd - "

message $\rightarrow$

1 1 0 1

□ Hamming Code, Parity Bit : $2^p \geq p + m + 1$

p = minimum parity bit required

m = number of message code

$$2^p \geq p + m + 1$$

$$2 + 4 + 1 = 7$$

$$2^3 \geq 3 + 4 + 1$$

$$8 \geq 8$$

$$p = 3$$

P_1	P_2	1	P_3	1	0	1
1	2	3	4	5	6	7

$$2^n = n = 0, 1, 2, 3$$

2^0	2^1	2^2
= 1	= 2	= 4

→ 2^0 position ચૂનાય
parity bit 3.

message →

1 1 0 1 0 - 5

$$2^n = n = 0, 1, 2, 3$$

2^0	2^1	2^2	2^3
= 1	= 2	= 4	= 8

$$P = 4$$

$$\text{संयुक्त संख्या} = 4 + 5$$

$$= 9$$

↓ position
↑ parity bit

P_1	P_2	1	P_3	1	0	1	P_4	0
-------	-------	---	-------	---	---	---	-------	---

1 2 3 4 5 6 7 8 9

• यदि message 5 बिट्स है, parity बिट्स 4 बिट्स.

	P_3	P_2	P_1	Even $(L + P) = 9$
0	0	0	0	
1	0	0	1	
2	0	1	0	
3	0	1	1	
4	1	0	0	
5	1	0	1	
6	1	1	0	
7	1	1	1	

$$P_1 \rightarrow 1, 3, 5, 7$$

$$P_2 \rightarrow 2, 3, 6, 7$$

$$P_3 \rightarrow 4, 5, 6, 7$$

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even parity হতে হবে,

$P_1 = 1$ হতে হবে

$P_2 = 0$

$P_3 = 0$

P_1	P_2	1	P_3	1	0	1
1	2	3	4	5	6	7



1	0	1	0	1	0	1
1	2	3	4	5	6	7

{ Parity bit গুলো
last-এ থেকেও
বরা যাবে }



Hamming code.

1010100 (ভুল হলে)

$$\text{Odd} \left\{ \begin{array}{l} P_1 \rightarrow 1110 \\ P_2 \rightarrow 0100 \\ P_3 \rightarrow 0100 \end{array} \right. \left| \begin{array}{l} \times \\ \times \\ \times \end{array} \right.$$

last-এ 0 হলে বা 00 হলে

even parity হবে না,

বি:দ্র: ভুল এটিগুলো ক্ষমাসুলভ দৃষ্টিতে দেখবেন ধন্যবাদ! 😊